

Gendered Governance and Environmental Attitudes: Does Political Agency Matter?

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Abstract

Can expanding women’s political empowerment shape citizens’ environmental attitudes and beliefs? We answer this using individual-level data from a broad sample of developed and emerging market economies over 1993–2020, employing instrumental variable and partial identification approaches. We find that women’s empowerment raises citizens’ environmental concern and heightens their sensitivity to environmental risk. However, it also reduces their personal willingness to bear the financial costs of protection. These effects hold similarly for men and women. We rule out the two channels most commonly proposed in the literature — policy stringency and institutional quality — as explanations for these effects on environmental concern and risk perception. Instead, our findings appear consistent with an agenda-setting mechanism associated with women’s empowerment. These attitude effects also translate into behavioural changes, including sorting recyclables, signing environmental petitions, and participating in environmental protests. By documenting a novel, policy-relevant determinant of citizens’ pro-environmental attitudes and behaviour, our study broadens the case for women’s empowerment beyond its direct policy effects. It highlights the role such empowerment can play in building the public support needed to sustain environmental protection over the long run.

Keywords: Women’s Empowerment, Pro-Environmental Attitudes, Pro-Environmental Behaviour, Agenda-Setting, Instrumental Variables, Partial Identification.

JEL Codes: Q53, Q54, J16, D72, C26, D91

1 Introduction

Environmental degradation is taking a heavy toll on human well-being. Air pollution alone is linked to roughly 7 million premature deaths each year (World Health Organisation), and 2024 was the warmest year on record globally, 0.12°C hotter than 2023 (Copernicus Climate Change Service, 2025). These figures underscore how high the stakes are. Governments have responded, but their efforts remain modest relative to the problem’s scale. Public funds made up \$197 billion of the \$220 billion spent worldwide on nature-based protection in 2023 — a small fraction of the \$7.3 trillion that went into nature-damaging activities that year (United Nations Environment Program, 2026). Yet effective environmental action depends not only on policy, but also on public support for that policy, without which such measures are harder to sustain (Chan, 2015; Ewald et al., 2022). Citizens appear broadly supportive: in a global survey of about 130,000 people across 125 countries, 86% said people in their country should be fighting global warming, and 69% said they would be willing to give up 1% of their household income every month towards that goal (Andre et al., 2024). Together, these facts motivate our focus on what shapes citizens’ environmental attitudes — a question central to addressing environmental degradation.

At the same time, women’s presence in politics has grown substantially. The share of women in national parliaments has risen from 11% in 1995 to 28% today, while their representation in local governments has grown from 20% in 2016 to 36% in 2025 (UN Women, 2026). This formal representation has been accompanied by a broader expansion in women’s political agency. Psychologists have long observed a gender gap in environmental attitudes: women consistently express more concern for the environment and are more likely to act sustainably than men (Xiao and McCright, 2015; Zelezny et al., 2000). Given this, a natural question arises: as more women enter political life, does their growing influence shift citizens’ environmental thinking?

This is an open empirical question. More specifically, we ask whether rising women’s political empowerment — captured not just by formal representation, but by women’s expanding capacity to make decisions in civil and political life — causally changes how

citizens think about and engage with environmental issues. Studying the demand side matters for several reasons beyond the policy-durability issue noted above. Citizen attitudes are the upstream force that may shape what politicians find it safe to endorse publicly in a democracy — even politicians personally committed to climate action may hold back if they doubt public support exists. However, politicians appear to misjudge that support systematically (Walgrave et al. 2023; Broockman & Skovron, 2018; Walgrave et al. 2022). This makes directly measuring citizen attitudes with survey data, as we do here, especially valuable for policymaking. Such attitudes also matter independently of policy, since they drive citizens’ everyday behaviours — recycling, consumption, support for local initiatives — that sustain environmental protection. To our knowledge, we provide the first causal evidence on an important determinant of people’s environmental preferences — women’s political empowerment. In doing so, we speak to how policy can affect citizens’ beliefs over the long run.

Empirically, answering this question involves three main challenges. First, individual-level data on environmental attitudes covering a large set of countries over time are scarce — most existing surveys lack the temporal coverage needed to study a slow-moving variable like women’s political empowerment. Second, isolating the causal effect of women’s empowerment on citizens’ attitudes is far from straightforward. Countries with more empowered women may also differ in economic development, institutional quality, and cultural values — all of which independently affect environmental preferences. Reverse causality compounds the problem — citizens who already hold stronger environmental attitudes may be more likely to bring about greater women’s empowerment in the first place. ¹ Because of this, a naive OLS estimate could be biased in either direction. Third, much of the existing literature relies on a narrow, one-dimensional measure of women’s political representation, which risks missing other channels through which empowerment changes citizen attitudes — potentially understating its true effect.

¹Though unrelated to environmental attitudes specifically, Van Der Pas et al. (2022) show that voters’ political ideology shapes their support for female versus male politicians, illustrating this channel’s broader plausibility.

Beyond the substantive research question, our work makes three contributions that address the challenges outlined above. First, we build a new cross-national dataset spanning 1993–2020 by combining rich individual-level data on environmental attitudes, behaviours, and sociodemographic characteristics from the ISSP with detailed country-level measures of economic development, structural characteristics, and political and institutional quality. This lets us study citizens’ attitudes directly, rather than relying on aggregate outcomes like air quality or CO_2 emissions, which capture only the policy-mediated consequences of women’s political empowerment. Politicians can move public opinion in other ways too — through what they choose to talk about publicly, or what their visibility in office signals to citizens — effects that need not show up in emissions or air quality data. Our approach therefore reflects the combined effect of these channels as citizens actually experience them, rather than a single policy-specific proxy.

Second, to move from correlation to causation, we instrument women’s political empowerment with gender quota adoption and estimate a TSLS model. Quotas shift women’s empowerment through a supply-side policy channel that plausibly does not directly affect citizens’ environmental attitudes. The main threat to this strategy is that quota adoption is not random — it may instead reflect a country’s broader movement towards post-materialist values— values that rise with income and industrialisation, and of which support for gender equality could be a component. Using data on the share of post-materialists in each country from the World Values Survey and European Values Survey, we find no evidence that post-materialism drives our results. We supplement this with a battery of further checks — formal and informal tests of the exclusion restriction, a rich set of individual and country-level covariates and country and year fixed effects, and Oster’s (2019) partial identification bounds on the role that unobserved confounders would need to play to overturn our findings (full details in Section 7). We also probe why the main result holds, using mediation analysis to test candidate mechanisms and rule out alternatives, and examine heterogeneity by gender to identify which citizens are most affected.

Third, rather than relying on a single measure of women’s parliamentary representation, our OLS analysis uses a multidimensional index of women’s political empowerment from

V-Dem. While this index has been used in other empirical contexts (Dahlum et al., 2022; Asongu et al., 2022), its application to the study of pro-social attitudes remains limited. It is constructed with careful attention to measurement error and data quality.

Turning to our baseline results, we find that greater women’s empowerment makes citizens more aware of environmental problems, more concerned about them, and more sensitive to the dangers posed by specific hazards. Yet it also makes them less willing to bear the financial costs of protecting the environment personally. In terms of magnitude, a one-standard-deviation increase in the empowerment index raises the ‘Environmental Concern Index’ by 0.22 standard deviations and the ‘Environmental Risk Perception Index’ by 0.17 standard deviations. It lowers the ‘Willingness to Pay Index’ by 0.17 standard deviations. The same patterns emerge when we look at the individual components of each index. One intuitive story ties these two sets of results together. Greater visibility of women in power appears to draw public attention towards environmental issues and lend them added credibility, making citizens more alert to the risks involved. This heightened awareness, however, does not carry over into a greater personal willingness to sacrifice. Since most countries in our sample are relatively developed and well-governed, citizens may simply trust empowered leaders to handle environmental problems on their behalf, lowering their own sense of responsibility to act personally. Notably, our mediation analysis rules out the two channels most commonly proposed in the literature – stricter environmental regulations and institutional quality – as explanations for the concern and risk perception effects specifically.

Our gender-based heterogeneity analysis reveals an interesting broader takeaway. Female political empowerment raises pro-environmental attitudes for men and women alike across every measure of environmental concern and risk perception, while dampening willingness to pay across both groups as well. We observe these results in both the baseline and instrumented specifications. Hence, our findings suggest that a gendered policy can have gender-neutral effects on environmental preferences of people, depending on how those

effects are transmitted to citizens.² This matters because a policy lever built around women’s representation and autonomy can still reshape public attitudes society-wide, as the underlying shift in norms diffuses across the population rather than staying confined to the group most explicitly targeted by the policy.

We also unpack which dimension of empowerment underlies these effects. Splitting the index into its three components points to distinct channels at work. Political participation shows effects broadly similar to the overall index for all attitude variables. Civil society participation also raises environmental concern and risk perception, plausibly through civic organisations that mobilise citizens directly – via grassroots campaigns, information sharing, and hazard-specific communication. Civil liberties matter for a narrower set of concern and risk perception outcomes: greater freedom of movement and economic independence let women observe environmental conditions firsthand, generally raising environmental concern and making local, proximate hazards like car pollution feel more dangerous. Civil liberties also reduce willingness to pay, consistent with households becoming more attentive to the financial costs of environmental protection. For every attitude variable, all three components move in the same direction throughout, suggesting these channels are complementary rather than substitutes.

To establish the causality of our findings, the identification strategy requires that our gender quota instrument satisfy the exclusion restriction. We carefully examine the various ways this could fail and show that none pose a major threat. Across all attitude variables, our TSLS estimates are significant, with a strong first stage and the same sign as OLS – though they are consistently larger in magnitude, pointing to downward bias in OLS. To illustrate, a one-standard-deviation increase in the instrumented index raises the perceived danger of car pollution by 18 percentage points. The effect is also economically meaningful – a country at the 75th percentile of women’s empowerment differs from one at the 25th percentile by 0.84 standard deviations; and our TSLS estimate implies that this gap alone corresponds to a 15 percentage point difference between the two countries in the probability

²The term ‘gendered policy’ refers to the policy’s design and not the policy’s goal.

of perceiving car pollution as a serious risk to the environment — compared to just 6 percentage points under OLS. Our gender quota instrument works mainly through the political participation component of the overall index — exactly the channel that quota laws are designed to affect — leaving the other two components largely unchanged.³

Our results continue to hold under a number of alternative IV estimators that do not depend on the exclusion restriction. We further carry out additional robustness checks and pre-adoption placebo tests, which show no evidence of reverse causality and support the exogeneity of the instrument.

One might reasonably worry that these attitudinal shifts are mere cheap talk. We show they are not — women’s empowerment also increases the probability that citizens engage in pro-environmental behaviours, including sorting recyclables, signing environmental petitions, and participating in environmental protests. Effects on avoiding harmful products and joining environmental groups are positive, though less precisely estimated. Together, these findings suggest that women’s empowerment may partially narrow the attitude-behaviour gap widely documented in the environmental psychology literature, with implications for building broader societal support for sustainable practices.

Complementing the existing supply-side literature on female political representation and environmental outcomes (Mavisakalyan and Tarverdi, 2019a; Rios et al., 2024; Ergas and York, 2012), we uncover nuanced demand-side effects that vary across domains of people’s environmental attitudes — revealing both the potential and limits of empowerment as a lever for environmental change. This matters at a moment when economic growth and environmental protection are so often framed as competing goals worldwide, and where shifts in public support can pull that trade-off towards better outcomes for the planet. For policymakers, practitioners, and researchers, our results indicate that citizens’ attitudes and behaviours are central outcomes to consider in understanding how environmental policy gets adopted and sustained. One policy-relevant way to influence these outcomes is to integrate women into high-status positions and expand their decision-making authority, which may

³Instrumenting each sub-component separately, we find that only the first stage F statistic for political participation exceeds the conventional weak instrument threshold of 10 (results available on request).

contribute meaningfully to environmental preservation — among the most urgent priorities the world faces today.

The remainder of the paper is organised as follows. Section 2 reviews the literature, while Section 3 develops the conceptual framework. Section 4 describes the data and Section 5 outlines the empirical strategy. Section 6 presents the baseline results, and Section 7 addresses endogeneity through a TSLS approach alongside additional specification and robustness checks. Section 8 examines pro-environmental behaviours, Section 9 discusses mediation analysis, and Section 10 concludes.

2 Literature Review

A well-established literature links gender of an individual to environmental attitudes: women display greater environmental concern and are more likely to act on it than men (Dietz et al., 2002; Hunter et al., 2004; McCright, 2010; Xiao and McCright, 2015; Vicente-Molina et al., 2018). This gender gap persists from childhood into adulthood (Zelezny et al., 2000). Two complementary theories help explain this gap. According to the gender socialisation hypothesis, women are raised to be more nurturing, cooperative, and caring, and show greater altruism towards the environment (Gilligan, 1993; Stern et al., 1993; Stern and Dietz, 1994). They also feel more responsible for protecting it. The safety concern hypothesis instead attributes the gap to gender differences in risk and health perceptions. Because of their caregiving nature, women are more sensitive to environmental hazards like pollution (Xiao and McCright, 2012). This raises their sense of urgency about long-term environmental degradation. Beyond these hypotheses, women are also disproportionately exposed to environmental harms (Birkmann et al., 2022), which shapes their perception of environmental problems and may drive their pro-environmental behaviour. Both theories are supported by empirical evidence. Women behave more cooperatively in managing shared resources (Tisserand et al., 2022), dominate grassroots environmental groups (Stoddart and Tindall, 2010), and show greater climate change awareness (McCright, 2010).

These gender differences are not only a demographic pattern — they may also carry

political relevance. Environmental attitudes and behaviour are not fixed. They are shaped by the broader political environment citizens are exposed to. One key feature of that environment is women’s political empowerment. Yet its role in shaping their environmental attitudes remains largely ignored in the literature.

A natural question that follows is whether this gender gap extends from citizens to political elites — whether female politicians themselves hold and act on stronger environmental preferences. The evidence is mixed. Fredriksson and Wang (2011) document that female legislators in the US House hold more pro- environmental views than their male colleagues, whereas the same pattern is not observed among Swedish parliamentarians (Sundstrom and McCright, 2014).

Where such gender differences in politicians’ preferences exist, they appear to translate into policy. Who holds office therefore matters for outcomes (Besley, 2005). Greater female representation improves health and education outcomes, raises foreign aid flows, and reduces corruption (Clots-Figueras, 2011, 2012; Hicks et al., 2016; Swamy et al., 2001; Brollo and Troiano, 2016). The environment is no exception. Higher female parliamentary representation leads to more environmental treaty ratifications, tougher climate policies, and reduced CO_2 emissions (Norgaard and York, 2005; Ergas and York, 2012; Mavisakalyan and Tarverdi, 2019a; Benlemlih et al., 2022). Similar effects are observed across a wide range of environmental outcomes — from clean energy transitions and green bond issuance to air quality, crop fires, and waste collection, and from forest governance to corporate boards (Gozgor et al., 2025; Al Mamun et al., 2024; Rios et al., 2024; Jagnani and Mahadevan, 2025; Baraldi et al., 2024; Agarwal, 2009; Kassinis et al., 2016; Orazalin and Baydauletov, 2020).⁴

These findings, however, mask meaningful variation across contexts — the influence of female politicians is not uniform. Tokenism remains a concern, with female leaders often seen as symbolic figures rather than substantive decision-makers in male-dominated settings (Green and Homroy, 2018; Torchia et al., 2011). Casarico et al. (2022) find that

⁴For a detailed review of this literature, see Hessami and Da Fonseca (2020).

mayor gender affects environmental spending only where councils also have a higher share of women. Jia et al. (2025) go one step further to point out that female representation improves environmental sustainability more where women have greater independent agency. This literature, throughout, examines policy outcomes. It says little about whether women’s political empowerment also shapes citizens’ own environmental attitudes and preferences — the question we take up. Since women’s presence in office does not automatically confer influence, we *also* study a broader concept of women’s political empowerment — capturing their capacity for choice, agency, and participation in decision-making — rather than descriptive representation alone. We explore the channels linking exposure to powerful women and citizens’ environmental attitudes in the next section.

3 Theoretical Mechanisms

We propose three mechanisms which can explain our findings. First, greater women’s political empowerment may change how citizens perceive environmental protection by altering its symbolic association with women as a political group — we call this the identity-signalling effect. This mechanism does not require citizens to observe or correctly track actual policies implemented by women leaders; it operates through the broader perception that women in power prioritize environmental issues. Consistent with this, female politicians are more likely than their male counterparts to support pro-environmental laws, and their rising representation is linked to more stringent climate policy (Ramstetter and Habersack, 2020; Mavisakalyan and Tarverdi, 2019a). As women gain political power, the issue of environmental preservation could become closely identified with them. Citizens tend to see women as well suited to speak on “care-based” issues — health, equality, and plausibly the environment (Huddy and Terkildsen, 1993; Eagly and Karau, 2002). This can lend the issue greater credibility, strengthening pro-environmental attitudes among some citizens.

For other citizens, however, the same association creates a risk of backlash. When women are seen as the group championing an issue, this can itself trigger distrust, independent of the issue’s content. Citizens tend to scrutinize women candidates’ competence more heavily

than similar men’s (Ditonto, Hamilton, and Redlawsk, 2014), so environmental messaging tied to women’s political leadership may be discounted on that basis alone. This effect can also outlast any single instance of female leadership. Cohen (2003) finds that once citizens learn which group supports a policy, they do not merely judge it differently — they come to perceive the policy itself differently, shaped by what they expect that group to stand for. If environmental protection becomes coded as a “women’s issue” more broadly, citizens who do not identify with that group may, by the same logic, downplay the severity of the underlying problem, weakening their pro-environmental attitudes.

Second, greater women’s political empowerment may increase the prominence of environmental issues in political debate and public discourse — we call this the agenda-setting effect. When particular environmental issues attract greater news attention, citizens tend to consider them as more important (McCombs and Shaw, 1972). This effect is strongest when citizens feel a high “need for orientation” on a topic — when the issue is relevant to them but they remain uncertain about it — prompting them to rely more heavily on the media to form their opinions (McCombs, 2001). This gives women’s empowerment a plausible route to shaping how visible environmental issues become, since women’s presence in politics appears to change what the media covers in the first place. Parties with more women MPs receive broader reporting on so-called “compassion issues” , while parties led by a woman are covered even more intensively on these issues (Greene and Luhiste, 2018). A similar dynamic appears further upstream, inside parties and legislatures: greater female representation among MPs and party leaders is associated with a stronger focus on social justice issues (Kittilson, 2011). Gender quotas also appear to create a “mandate effect” — new female legislators feel responsible for representing underrepresented groups’ interests, spurring greater legislative activism even where actual policy outcomes lag behind (Franceschet and Piscopo, 2008). Beyond making environmental issues more salient overall, this channel may also influence which particular aspects of the issue citizens notice. This is attribute agenda-setting — the media conveys not just an issue’s importance, but the salience of its specific characteristics (McCombs, 2015). If women’s empowerment reallocates coverage towards the dangers of environmental degradation, this could help explain the rise

in citizens' environmental concern and risk perception. Altogether, the evidence points to a chain: women's empowerment raises legislative and party attention to specific environmental topics, which in turn broadens and sharpens media coverage of those issues. As a result, citizens become more attuned to environmental harm.

However, greater media attention can cut both ways: the same coverage that raises awareness of environmental harm may also raise awareness of the economic costs of addressing it — higher energy prices or job losses in pollution-intensive industries, for example. Citizens who place greater weight on these costs may become less concerned about environmental protection as a result. This is not because they doubt the problem is real, but because the economic framing competes for their attention with the environmental one.⁵ This dampens their concern, distinct from willingness to act.

Third, even if citizens believe an environmental problem is serious, women's political empowerment may still reduce their willingness to act on that belief — we call this the political substitution mechanism. Believing a problem is serious is not the same as being willing to personally bear the cost of fixing it. If women leaders visibly take on that responsibility — through stricter policy and credible enforcement (Mavisakalyan and Tarverdi, 2019a) — citizens may reasonably conclude that their own contribution matters less at the margin. As a result, they come to trust empowered institutions to handle environmental protection, and addressing the problem becomes a collective task rather than a personal one. This logic follows from a well-established finding in public economics: when the government steps up provision of a public good, private contributions toward that same good tend to fall, as people treat collective and individual provision as substitutes (Bergstrom, Blume, and Varian, 1986; Andreoni and Payne, 2003). This channel weakens citizens' willingness to personally pay for safeguarding the environment.

Theory offers a competing prediction, however. As women's political empowerment

⁵For example, when carbon taxes are extensively reported by the media, some citizens may become more convinced of the need for environmental preservation, while others focus on the resulting increase in household energy bills. This illustrates how the same issue coverage can strengthen environmental concern among some citizens while weakening it among others.

raises the salience of environmental issues, it may also activate citizens' civic duty — the utility derived simply from complying with a social norm to contribute to collectively beneficial outcomes (Riker and Ordeshook, 1968) ; or instrumental altruism — that is, caring about how such outcomes affect others (Fowler, 2006; Edlin, Gelman, and Kaplan, 2007). Either motive would predict that citizens become more willing to personally contribute to environmental protection.

The combined effect of these mechanisms is ultimately an empirical question, giving rise to two testable hypotheses. The first concerns citizens' underlying beliefs about the environment — whether it is being harmed, and how severely. Because the net effect of the first two mechanisms is theoretically ambiguous, we cannot determine a priori the sign of the effect of women empowerment on environmental concern and risk perception. We therefore test the following hypothesis:

H1: Women's political empowerment affects citizens' environmental concern and risk perception.

The second hypothesis concerns a different question — not whether citizens believe the environment is at risk, but who they believe should bear the cost of addressing it. Here too, the direction of the net effect of the third mechanism cannot be established on theoretical grounds alone. This motivates our second hypothesis.

H2: Women's political empowerment affects citizens' willingness to pay for environmental protection.

4 Data

We build an individual-country-year dataset from a range of sources. In this section, we describe the data sources, variable construction, and summary statistics.

4.1 Measuring pro-environmental attitudes

Our main data source for the pro-environmental attitudes variables is the International Social Survey Programme (ISSP) Environment modules. The Environment modules cover attitudes towards environmental issues, environmental behaviours, willingness to pay for environmental protection, and preferences regarding government environmental measures. We construct our pro-environmental attitudes variables from the four rounds in which the ISSP fielded an Environment module: 1993, 2000, 2010, and 2020. We prefer the ISSP over the Integrated Values Survey (IVS) despite the latter’s broader country coverage, for two reasons. First, the IVS lacks consistent availability of pro-environmental attitude variables across rounds, whereas the ISSP Environment modules offer a stable set of measures repeated over multiple rounds. Second, the ISSP Environment modules cover a richer set of environmental attitude and behaviour variables, compared to the IVS.

Self-reported attitude data have well-known limitations, including measurement error. Three features of our setting are worth noting in this regard. First, attitudes are the direct object of interest for our research question, rather than a proxy for some other outcome — so even if measured with some noise, they are not a poor substitute for what we actually want to study. Second, our results are robust across a wide range of specifications and estimation methods, which would be unlikely if they were purely due to noisy self-reports. Third, our conclusions extend to pro-environmental behaviours as well, providing some corroborating evidence beyond attitudes alone. While these too are self-reported, the fact that two distinct measures — beliefs and actions — move in the same direction in a theoretically consistent way offers some reassurance against our findings being driven by noisy self-reports.

The ISSP Environment module measures attitudes on five-point scales, though the anchors vary across questions. We construct two sets of outcome variables: i) binary indicators that take the value one if the response reflects a pro-environmental attitude and zero otherwise, and ii) ordinal measures that retain the original five-point scale to preserve the ordering and intensity of responses. Table 1 describes how we derive the binary

indicators of pro-environmental attitudes from the original variables.

When constructing the ordinal outcome variables, we reverse-code those variables whose original scale runs in the opposite direction — that is, where higher numeric values indicate weaker, rather than stronger, pro-environmental attitudes. This applies to Modern Life Harms Environment, Willingness to Pay Higher Prices, Willingness to Pay Higher Taxes, Willingness to Accept Lower Living Standards, Costly Environmental Action, and all five risk perception variables. After reverse-coding, higher values consistently indicate stronger pro-environmental attitudes across all measures.⁶

To address multiple hypothesis testing concerns, we construct three summary indices following Anderson (2008): an environmental concern index, a willingness to pay index, and an environmental risk perception index.⁷ Each index is constructed as an inverse-covariance weighted average of the underlying survey responses and standardised to have a mean of zero and a variance of one. The ‘Environmental Concern Index’ (ECI) aggregates three variables: Future Environment vs. Current Jobs, Modern Life Harms Environment, and Progress Harms Environment. The ‘Willingness to Pay Index’ (WTPI) aggregates three variables: Willingness to Pay Higher Prices, Willingness to Pay Higher Taxes, and Willingness to Accept Lower Living Standards. The ‘Environmental Risk Perception Index’ (ERPI) aggregates five variables: Risk of Car Pollution, Risk of Industrial Air Pollution, Risk of Agricultural Chemicals, Risk of Water Pollution, and Risk of Global Temperature Rise.⁸ In addition to Anderson indices, we also present results using a simple average of the underlying variables and principal component analysis as alternative aggregation methods.

Our basic analysis of the three indices reveals several intuitive patterns. Spain exhibits

⁶We exclude two questions from our analysis - ‘Economic Growth Harms Environment’ and ‘Nuclear Power Stations are Dangerous For Environment’ - because neither can be coded unambiguously as reflecting a pro-environmental attitude.

⁷We construct the Anderson(2008) index using the Stata module `swindex`, developed by Schwab et al. (2020). This approach has been used in several recent studies to build indices across different domains (Dhar et al., 2022; Allcott et al., 2020; Roth et al., 2022).

⁸We exclude Personal Environmental Agency and Costly Environmental Action from the indices as they do not fit cleanly with the other components and would compromise the conceptual clarity of the indices.

the highest mean environmental concern (mean = 0.226), while the Philippines shows the lowest (mean = -0.442). For willingness to pay for environmental protection, Switzerland displays the highest mean (mean = 0.466) whereas Latvia shows the lowest (mean = -0.689). Chile has the highest mean environmental risk perception (mean = 0.710), and the Netherlands the lowest (mean = -0.587).

4.2 Measuring female empowerment

Our treatment variable is the Women’s Political Empowerment Index from the Varieties of Democracy (V-Dem) project. Following Sundstrom et al. (2017), we define women’s political empowerment as a process of increasing capacity for women, leading to greater choice, agency, and participation in societal decision-making. The index is multi-dimensional — it captures not only women’s representation in formal political roles but also their civil liberties and participation in civil society. We prefer it over a single measure of women’s parliamentary representation for this reason.⁹ Specifically, it is constructed as the weighted average of three sub-indices: women’s civil liberties, women’s civil society participation, and women’s political participation. The index ranges from 0 to 1, with higher values indicating greater empowerment.

The index varies substantially across countries in our sample. Figure A1 shows the variation in this index across countries over our sample period. Denmark has the highest mean women’s political empowerment (mean = 0.96) and Russia the lowest (mean = 0.65).

¹⁰

We compile data on our instrument - gender quota adoption - from multiple sources. Country-level data on the year of quota adoption through 2005 comes from Dahlerup (2005). We extend this coverage beyond 2005 using two additional sources: the Global Database for Quotas for Women, maintained by the Inter-Parliamentary Union and the Quota Adoption

⁹Our results are robust to alternative measures of women’s political empowerment, including the share of women in parliament.

¹⁰These means pertain to the non-standardised index, which ranges from 0 to 1. The index shows substantial variation across country-years as well as within-country variation over time.

and Reform Over Time dataset of Hughes et al. (2017).

4.3 Covariates

We source GDP — expressed in constant 2021 international dollars adjusted by purchasing power parities (PPPs) — from the World Bank World Development Indicators. Population figures come from the UN World Population Prospects 2024, measured as of 1st July. Several other variables — trade openness, population density, the Ease of Doing Business ranking, and the World Governance Indicators — are all drawn from the World Bank. Trade openness is defined as the sum of exports and imports of goods and services as a share of GDP, expressed in percent. Population density is measured as population per square kilometer of land area. The Ease of Doing Business rank is determined by aggregating scores across 10 equally weighted topics — starting a business, dealing with construction permits, getting electricity, registering property, getting credit, protecting minority investors, paying taxes, trading across borders, enforcing contracts and resolving insolvency. Countries are assigned a rank from 1 to 190, with a lower rank number (i.e., a rank closer to 1) indicating a more business-friendly regulatory environment. The governance indicators (Control of Corruption and Government Effectiveness) are each scored on a 0–100 scale, where higher values reflect better governance. We obtain the HDI from the UNDP Human Development Report. It captures achievement across three dimensions — health, knowledge, and living standards — computed as the geometric mean of normalised indices for each dimension. We draw sectoral employment shares in agriculture, industry, and services from the World Bank, with each defined as a percentage of total employment.

Natural resource wealth — measured as the value of oil and gas net exports per capita — comes from Ross and Mahdavi (2015). We obtain the political orientation of the chief executive’s party (left, right, or center) from the Database of Political Institutions (DPI), maintained by the Inter-American Development Bank. Finally, we source individual-level sociodemographic characteristics from the ISSP. Table 2 consolidates the data sources and presents summary statistics for these covariates.

5 Empirical Strategy

We estimate the following specification using OLS to assess the baseline relationship between women’s political empowerment and citizens’ pro-environmental attitudes:

$$Y_{ict} = \alpha_0 + \beta_1 WPEI_{c,t-1} + \beta_2 X_{ict} + \beta_3 Z_{c(i),t-1} + \gamma_{c(i)} + \delta_t + \epsilon_{ict} \quad (1)$$

where Y_{ict} denotes a pro-environmental attitude variable or an index combining several such variables for individual i in country c at time t . $WPEI_{c,t-1}$ is the Women’s Political Empowerment Index in country c at time $t-1$ obtained from V-Dem. This index is standardised to have a mean of zero and a standard deviation of one so that β_1 can be interpreted in standard deviation units. X_{ict} is a vector of time-invariant or near time-invariant individual-level controls including age, sex, education and place of living. $Z_{c(i),t-1}$ is a vector of country-level controls such as log GDP per capita, sectoral employment shares, trade openness, Control of Corruption, Government Effectiveness, Doing Business Rank, polity score, log population density, Chief Executive party orientation, and oil and gas net exports per capita. We lag the treatment variable and country-level controls by one year, following Lin and Chu (2026). This is also motivated by a theoretical consideration – women’s political empowerment is likely to affect citizen environmental attitudes with a delay, as attitude formation takes time.¹¹ Country fixed effects $\gamma_{c(i)}$ absorb all time-invariant country characteristics that may influence environmental attitudes. We also control for year fixed effects that partial out shocks to environmental attitudes which are common across countries but vary over time – for example, global economic conditions or major international climate events that shift public attention to environmental issues.

¹¹As mentioned in the text, we lag the treatment and country-level controls by one year relative to the attitude variables. For example, attitudes measured in 1993 are matched with treatment and controls from 1992, and attitudes measured in 2000 are matched with treatment and controls from 1999. As an alternative, we address the bad controls problem more directly by using baseline values of time-varying controls interacted with a linear time trend as well as year dummies instead of lagged controls (Angrist and Pischke, 2009). This approach yields coefficient estimates that are consistent with our main results (results not reported).

In alternative specifications, we replace year fixed effects with region-year fixed effects to additionally absorb time-varying shocks common to countries within the same region. All specifications are weighted using the survey weights provided in the ISSP. Standard errors are computed using the wild cluster bootstrap, clustered at the country level, which allows for within-country correlation of shocks to environmental attitudes over time. This approach provides more reliable inference than conventional cluster-robust standard errors, given the small number of clusters (36). Our coefficient of interest is β_1 , which captures the within-country association between women’s political empowerment and people’s pro-environmental attitudes, net of individual controls, country-level controls, and fixed effects.

We select control variables based on their potential to affect either women’s political empowerment or pro-environmental attitudes; omitting them would otherwise bias our estimates. We include controls for individual characteristics, economic development, structural characteristics of the economy, and political and institutional quality. Below, we provide justification for including each covariate.

Among individual-level controls, we incorporate age since environmental concern and risk perceptions vary systematically across age groups (McCright, 2010), and additionally control for age-squared to allow for a possible non-linear relationship (Geys et al., 2021). We control for gender, reflecting evidence that women tend to express more favourable environmental attitudes than men (Dietz et al., 2002; Zelezny et al., 2000; McCright, 2010). This individual-level gender gap is conceptually distinct from the country-level women’s political empowerment index, which is our main variable of interest. Education affects environmental awareness and concern (McCright, 2010), so we control for it as well. Place of living matters because urban and rural residents have different environmental exposures and likely hold different environmental attitudes (OECD, 2011).

Our first set of country-level controls captures economic development. GDP per capita is the strongest cross-national predictor of women’s political empowerment (Dahlum et al., 2022) and richer countries tend to exhibit greater environmental concern along with stronger institutional capacity for environmental governance (Franzen and Meyer, 2010), making it an important potential confounder. Population density is a related control —

densely populated areas may be exposed to higher pollution and environmental health risks, which can raise environmental risk perceptions and concern (Franzen and Meyer, 2010).

Our second set of controls relates to the structural characteristics of the economy. Sectoral composition matters — industrial economies typically produce more pollution, affecting citizens’ risk perceptions, while citizens in agricultural economies rely directly on the environment for their livelihoods, which influences their attitudes towards it. This same industrial or agricultural orientation can also shape women’s electoral prospects (Rios et al., 2024). Trade openness enters the specification for two reasons: more open economies are exposed to international environmental norms and standards that may heighten environmental awareness and affect citizens’ risk perceptions. It has been also been shown to promote women’s political rights across countries (Neumayer and De Soysa, 2011). We also include oil and gas net exports per capita, as countries that rely more heavily on these exports tend to have lower female political participation (Mavisakalyan and Tarverdi, 2019b) and may hold distinct environmental attitudes due to their dependence on fossil fuels.

Our third set of controls addresses political and institutional quality. Citizens living under democratic rule have greater scope to express their environmental preferences than those under autocratic regimes (Farzin and Bond, 2006). Moreover, democracies have generally adopted stricter environmental policies than non-democracies. Both factors may shape citizens’ environmental concern. More democratic countries also tend to have higher female representation (Kim, 2019), making polity score — our proxy for democracy — a potential confounder. We also control for the orientation of the chief executive’s party, because left-leaning governments tend to favour both environmental preservation and gender equality. Without this control, our women’s political empowerment variable may instead pick up the effects of ruling party ideology. This possibility is reinforced by evidence that left-leaning parties tend to adopt a more pro-environmental stance (Neumayer, 2004). Turning to institutional quality, corruption can undermine the quality of environmental governance and the credibility of institutions, both of which can shape citizens’ environmental attitudes. Government effectiveness plays a similar role, as citizens who trust institutions to address

environmental problems are likely to hold different levels of environmental concern. Because female empowerment is also linked to better institutional quality overall (Dollar et al., 2001; Swamy et al., 2001), both corruption and government effectiveness are potential confounders. Finally, our results may simply reflect a broader tendency for countries with higher women’s political empowerment to have more extensive policymaking and stronger regulatory environments overall. Following Mavisakalyan and Tarverdi (2019a), we account for this possibility by adding countries’ Doing Business ranks as a proxy for general regulatory activity. ¹²

6 Results

6.1 Baseline results

Table 3 presents the OLS estimates of equation 1. Recall that, as discussed in the data section, each attitude variable is coded as binary, equal to 1 where it reflects a pro-environmental stance. Column 2 reports a parsimonious specification with no controls, in which women’s political empowerment shows no significant effect on most of the pro-environmental attitude variables. Once we introduce relevant individual- and country-level controls in column 3, however, all attitude variables exhibit statistically significant and economically meaningful effects, while the signs of the estimates remain unchanged. We consider column 3 as our preferred specification, since controlling for these covariates — which determine women’s empowerment or environmental attitudes — nets out their effects. This yields more precise estimates and mitigates concerns about omitted variable bias.

We first consider outcomes broadly classified as measures of abstract environmental concern or belief. A one-standard- deviation increase in the women’s empowerment index raises the probability of holding a pro-environmental attitude towards ‘Future

¹²Note that, for some outcomes, certain country-level institutional and political controls are highly correlated with each other. In such cases, we retain only a subset of these variables, ensuring that the Variance Inflation Factor (VIF) of each covariate remains below 10, and there is no serious multicollinearity issue.

Environment vs. Current Jobs’ by 4.6 percentage points; this variable captures how much weight people place on environmental protection relative to economic concerns. The same increase raises the probability of a pro-environmental attitude towards ‘Modern Life Harms Environment’ and ‘Progress Harms Environment’ by 4.2 and 6.8 percentage points, respectively – both variables capturing beliefs about the existence and severity of environmental problems. Finally, it raises the probability of a pro-environmental attitude towards ‘Personal Environmental Agency’ by 6.4 percentage points, a variable measuring whether people believe that environmental action is both possible and worthwhile.

We next turn to outcomes reflecting willingness to pay or sacrifice for the environment. Here, the pattern is different: A one-standard-deviation increase in the index lowers the probability of a pro-environmental attitude towards ‘Willingness to Pay Higher Prices’, ‘Willingness to Pay Higher Taxes’, and ‘Willingness to Accept Lower Living Standards’ by 11.6 , 10.8 and 10 percentage points, respectively. The index also lowers the probability of a pro-environmental attitude towards ‘Costly Environmental Action’ by 15.3 percentage points – a variable capturing individuals’ willingness to personally bear the costs of protecting the environment.

The final set of outcomes relates to environmental risk perception. A one-standard-deviation increase in the index raises the probability of a pro-environmental response to ‘Risk of Car Pollution’, ‘Risk of Industrial Air Pollution’, ‘Risk of Agricultural Chemicals’, ‘Risk of Water Pollution’, and ‘Risk of Global Temperature Rise’ by 6.6, 8.9 , 17.6 , 5.7 and 12.3 percentage points, respectively. The effects on ‘Risk of Agricultural Chemicals’ and ‘Risk of Global Temperature Rise’ stand out as especially strong: they hold even unconditionally, before any controls are added to the specification. The direction of the omitted variable bias differs between the two, which is informative: for agricultural chemicals, the estimate rises once controls are added, suggesting that the omitted variables bias the coefficient downward on average; for global temperature rise, the estimate falls on incorporating controls, implying that the omitted variables bias the coefficient upward on average. This divergence suggests that no single dominant confounder is uniformly driving our results across outcomes.

Taken together, these findings indicate that women’s political empowerment shifts citizens’ environmental attitudes along several dimensions. It raises citizens’ environmental concern — their sense that the environment is under threat — while also elevating their risk perception, leading them to view particular hazards like air and water pollution and global warming as truly dangerous. One plausible explanation is that female leaders bring environmental issues greater public attention and credibility, making related risks more salient and memorable to citizens.¹³ This heightened awareness, however, does not translate into greater personal willingness to sacrifice: if anything, empowerment reduces citizens’ expressed willingness to pay higher prices or taxes, or to accept lower living standards, for environmental protection. As a large share of countries in our sample are developed with strong governance, citizens may reasonably trust empowered female leaders to tackle environmental problems on their behalf, reducing their motivation to personally contribute. These findings align with the existing evidence that women’s political representation reduces measured environmental harms, such as air pollution and CO_2 emissions (Mavisakalyan and Tarverdi, 2019a; Rios et al, 2024). The co-occurrence of heightened risk perception and improved environmental outcomes under women’s empowerment suggests that these effects are complementary rather than substitutes, though we do not investigate this relationship directly here.

To avoid losing information through binarizing the outcome variables, we also estimate ordered logit models using the original five-category ordinal measures. As described in the data section, these ordinal variables are coded so that higher values correspond to stronger pro-environmental attitudes. This approach has an added advantage: it allows us to capture how women’s empowerment affects the intensity of pro-environmental attitudes, rather than simply whether such attitudes are present.¹⁴

¹³For instance, environmental policies promoted by female leaders can attract substantial legislative and media coverage. On June 2, 2026, the House passed Representative Alexandria Ocasio-Cortez’s Geothermal Cost-Recovery Authority Act as part of the broader Geothermal Energy Advancement Act, aimed at accelerating geothermal energy development. See https://docs.house.gov/billsthisweek/20260601/H5631_SUS_xml.pdf for the bill text.

¹⁴The Brant, Likelihood-Ratio and Wald tests confirm that the proportional odds assumption is not

Table A1 presents the ordered logit estimates for the thirteen outcome variables, each measured on a five-point scale. The results are consistent with those in Table 3. Among the environmental concern and belief variables, the largest effect is for ‘Future Environment vs. Current Jobs’: a one-standard-deviation increase in the women’s empowerment index raises the odds of expressing a more pro-environmental attitude by 48%. This is intuitive, reflecting people’s strong prioritization of long-term environmental protection over short-term economic outcomes. Among the willingness to pay or sacrifice variables, the largest effect emerges for ‘Costly Environmental Action’, which lowers the odds of a more pro-environmental attitude by 30%. This is plausible, since this variable sets a higher bar than the other willingness-to-pay variables: rather than asking about a hypothetical willingness to pay higher prices or accept a lower standard of living, it asks whether respondents already act on environmental principles despite the associated time and monetary costs. Among the risk perception variables, we observe the strongest effects for ‘Risk of Car Pollution’ and ‘Risk of Global Temperature Rise’, both raising the odds of a more pro-environmental attitude by 29%. This likely points to the severity and salience of air pollution and global warming as two of the most pressing global environmental concerns.

Table 4 aggregates results across related outcomes using three summary indices, addressing concerns around multiple hypothesis testing. Following Anderson (2008), a one-standard-deviation increase in the women’s empowerment index is associated with a 0.22 standard deviation increase in the ECI, a 0.17 standard deviation decrease in the WTPI, and a 0.17 standard deviation increase in the ERPI. These results are robust to alternative index construction methods, including Principal Component Analysis (columns 4 and 5) and simple averaging (columns 6 and 7). Overall, both the OLS and ordered logit estimates tell a consistent story: women’s empowerment strengthens pro-environmental attitudes only where doing so imposes no cost on the individual — abstract beliefs and concerns about the environment, and perceptions of environmental risk — while weakening the corresponding attitudes tied to willingness to pay or sacrifice. These latter variables ask respondents

violated for our ordered logit specifications, supporting the use of the ordered logit model over alternative specifications. Full results are available on request.

to state a hypothetical willingness to bear a personal cost or report already bearing one, which is fundamentally different from simply holding a belief or judging something to be environmentally risky.

6.2 Components of the Women Political Empowerment Index

The Women’s Political Empowerment Index, obtained from V-Dem, is the weighted average of three sub-indices: women’s civil liberties, women’s civil society participation, and women’s political participation. The Political Participation Index (PPI) indicates the degree to which women are descriptively represented in formal political positions. The Civil Society Participation Index (CSPI) captures women’s ability to voice their views and to form and take part in groups. The Civil Liberties Index (CLI) denotes women’s capacity to make meaningful decisions in key areas of their lives. In this section, we examine how each of these component indices affects pro-environmental attitudes, to better understand the mechanisms underlying our baseline results.

Table 5 reports estimates using each of the three WPEI sub-components as a separate treatment variable. When the PPI is used as the treatment, the signs of the coefficients across all thirteen pro-environmental attitude variables match those in Table 3, suggesting that the same mechanisms discussed earlier are likely at work here as well.

For the environmental concern and belief outcomes, women’s civil society participation plays a prominent role in the improvement of pro-environmental attitudes. This makes sense -- women’s civic organizations mobilize environmental concern directly, through grassroots campaigns, information sharing, and collective advocacy.¹⁵ Women’s civil liberties, by

¹⁵Examples abound. Women were at the forefront of India’s Chipko movement against deforestation. There are numerous other women activists across both developed and developing countries who have championed environmental causes, including Greta Thunberg, Maggie White, and Lois Gibbs (<https://gendermatters.in/40-women-environmentalists/>). Women have played a prominent leadership role in environmental activism, particularly through local social movement organizations (Breton, 1998). Such leadership has included organizing against industrial pollution (Cable, 1992), forest protection movements such as Chipko and the Green Belt Movement (Zelezny and Bailey, 2006) -- often drawing significant public

contrast, matter for a narrower set of outcomes — specifically, ‘Progress Harms Environment’ and ‘Personal Environmental Agency’. Greater civil liberties expand women’s freedom of movement and economic independence, allowing them to observe environmental conditions firsthand and to seek justice when environmental harms occur. This, in turn, raises the visibility of environmental problems along two channels. First, more direct exposure to environmental conditions may heighten public awareness of environmental degradation over time, strengthening attitudes towards ‘Progress Harms Environment’. Second, seeing environmental harms get successfully redressed shows citizens that environmental action is both possible and effective, reinforcing their sense of agency and raising ‘Personal Environmental Agency’.

For the willingness to pay outcomes, women’s civil liberties, however, reduce citizens’ willingness to personally sacrifice for the environment. Greater civil liberties are linked to stronger property rights and greater freedom from forced labour, both of which make women more economically empowered. As women become more influential in household financial decisions, households collectively may become more attentive to the financial costs of environmental action — higher prices, taxes, or reduced living standards — lowering people’s overall willingness to pay for environmental protection. By contrast, women’s civil society participation leaves willingness to pay unaffected, despite increasing environmental awareness and risk perception among citizens.

Women’s civil society participation also shapes citizens’ environmental risk perceptions, with the largest effect observed for global temperature rise. One plausible channel is that women-led environmental NGOs may communicate specific hazard risks to citizens — for example, through pollution monitoring data, public health advisories, or documented cases of environmental harm — making these risks more salient than the broader framing used to build environmental concern. Additionally, because global warming affects entire communities rather than select individuals, it may feel more severe to citizens. In contrast, women’s civil liberties makes people more likely to perceive local, proximate hazards —

attention to environmental issues.

car pollution, industrial air pollution, and agricultural chemicals — as dangerous, but not global warming. This pattern is consistent with the idea that greater freedom of movement exposes women more directly to these local hazards, raising public awareness of proximate environmental risks more broadly.

Overall, the three sub-indices move in the same direction across all outcomes, increasing our confidence that the results are not driven by confounding factors. While it is not feasible to cleanly separate the effect of each sub-index, the nuanced pattern across dimensions suggests that women’s empowerment operates through multiple, interrelated channels to alter environmental attitudes.

7 Addressing Endogeneity

7.1 Using an instrument

Our findings so far show a robust positive relationship between women’s political empowerment and pro-environmental attitudes in the concern and risk perception domains, and an equally robust negative relationship in the willingness to pay domain. These relationships, however, are not causal. Women’s political empowerment is not randomly assigned across countries, and countries with higher levels of empowerment may differ in other unobserved ways that independently affect environmental attitudes. While our earlier specifications control for a rich set of observed confounders, unobserved heterogeneity can never be fully ruled out. Reverse causality is an additional concern. One solution to these problems is to identify a source of exogenous variation in women’s empowerment — an instrumental variable — and estimate the relationship using two-stage least squares (TSLS). In what follows, we justify our choice of instrument, explain its validity, and present the resulting TSLS estimates for our attitude outcomes.

We use the adoption of gender quotas as an instrument for women’s political empowerment. A gender quota requires that a specified share of positions on a list, in a group, or within an institution be held by women (Clayton & Zetterberg, 2018; Hughes et al., 2019). Gender

quotas are of three broad types. Reserved seat quotas set aside a fixed proportion of seats for women, regardless of how many women contest. Candidate quotas require parties to nominate a minimum proportion of women, without guaranteeing that they are elected. Voluntary party quotas differ from both: they exist when individual parties commit, on their own initiative rather than under legal obligation, to nominate a certain share of female candidates on their electoral lists. Countries adopting gender quotas have been shown to achieve significantly higher representation of women in both local government and national parliaments (Dimitrova-Grajzl & Obasanjo, 2019).

Gender quotas function as a supply-side policy intervention that directly increases female political representation by setting minimum thresholds. This makes the exclusion restriction more plausible: quota adoption is likely to affect citizens' pro-environmental attitudes only through its effect on women's political empowerment. In our context, we believe quota adoption is a better instrument than years since female suffrage, which is the instrument used by Mavisakalyan and Tarverdi (2019a), or than any other intervention that shifts citizens' demand curve directly.¹⁶ Our instrument, quota adoption, is coded as a binary variable. It equals 1 for all years after a country adopts a quota, and 0 otherwise - covering both the pre-adoption period for adopting countries and the entire sample period for non-adopting countries.

Even so, quotas are not adopted arbitrarily — they are implemented for specific political, economic and social reasons, and it is worth considering the channels through which quota adoption could violate the exclusion restriction. The literature identifies several broad reasons why countries introduce gender quotas: strategic electoral incentives among political elites, evolving domestic norms around gender equality, and influence from international organisations, treaties or policy diffusion from neighboring countries.¹⁷ A further, more

¹⁶Years since suffrage may directly affect citizen attitudes through several cultural and developmental channels that are difficult to account for. Beyond this exclusion restriction concern, using years since suffrage as an instrument for women's political empowerment also yields insignificant first-stage coefficients for some outcomes, along with weak first-stage F-statistics.

¹⁷For details, see Krook (2006) and Weeks (2018). Another plausible reason is organised advocacy by women's movements which likely overlaps with the civil society participation component of our treatment

distinct concern is reverse causality— citizens who already hold stronger pro-environmental preferences may be more likely to elect and empower women, including through quota adoption, independent of any effect running from empowerment to attitudes.

We respond to these concerns in the following ways. One possibility is that a country’s quota adoption is linked to a broader shift towards progressive, postmaterialistic values — values that tend to rise with industrialisation and income, and of which support for gender equality may be a part. We test this directly, controlling for a country-level measure of postmaterialism, that is, the share of postmaterialist individuals in a country, constructed from the European Values Study (EVS) and World Values Survey (WVS). Table 6 reports the results. With a strong first stage, our TSLS estimates including this control are consistent with the baseline results without it. We also find that the gender quota instrument does not significantly predict the postmaterialist share, with a coefficient of -0.0016 ($SE=0.0236$). This suggests that quota timing is not simply tracking a broader post materialist trend, and instead operates through women’s empowerment specifically. The remaining two channels — elite strategic incentives and influence from international organisations or regional policy diffusion — are harder to rule out directly. But to the extent that they act similarly across treated and control countries or are absorbed by our time varying controls and fixed effects, they should not bias our estimates. Our results are also robust to a range of alternative IV estimators that do not rely on the exclusion restriction at all (see section 7.6). Finally, we address the reverse causality threat with a pre-adoption placebo test (see section 7.4) This test asks whether future quota adopters already exhibit different environmental attitudes before the quota takes effect. We find no such pattern — the estimates are small and statistically insignificant, supporting our identification strategy.

To further show the validity of our instrument, we conduct an informal check of whether the exclusion restriction is plausible. Column (2) of Table 7 regresses each outcome directly on the gender quota instrument, without controlling for women’s political empowerment. Here, the coefficient on gender quota is positive and significant for the environmental variable, and does not reflect a confounding factor.

concern/belief and risk perception variables, and negative and significant for the willingness to pay/sacrifice variables. Once we add the empowerment control in Column (3), however, the coefficient on gender quota becomes insignificant – consistent with the quota affecting these outcomes primarily through women’s empowerment, rather than through some independent direct channel.

With this suggestive evidence in support of the exclusion restriction, we now examine the first stage. To do so, we treat women’s political empowerment as endogenous and instrument it using country-level gender quota adoption. The first stage specification is:

$$WPEI_{c,t-1} = \alpha_1 + \theta_1 GQ_{c,t-1} + \theta_2 Z_{c,t-1} + \lambda_c + \eta_{t-1} + u_{c,t-1} \quad (2)$$

$WPEI_{c,t-1}$ denotes the treatment variable – Women’s Political Empowerment Index for country c at time $t-1$. $GQ_{c,t-1}$ is a dummy variable taking the value 1 for adopting countries in their post-adoption years and 0 otherwise.¹⁸ All other variables follow the same definitions as in equation 1. Our coefficient of interest is θ_1 , which captures the effect of gender quota adoption on women’s political empowerment.¹⁹

Tables 8 and A2 report our first-stage results for the individual attitude variables and summary indices, respectively, further reinforcing our instrument’s validity. We find a positive and significant relationship between gender quota adoption and women’s political empowerment.²⁰ Although few countries in our sample have adopted a gender quota, the first stage remains sufficiently powered to detect the effect. The Kleibergen-Paap (2006) F-statistics for the weak identification test substantially exceed the conventional rule-of-thumb threshold of 10 in all cases. This suggests that our instrument is neither weak nor

¹⁸Our gender quota variable covers legislated candidate quotas and reserved seats for women across national, regional, and local governments.

¹⁹Our second stage regression specification is as follows: $Y_{ict} = \alpha_2 + \psi_2 \widehat{WPEI}_{c,t-1} + \psi_3 X_{ict} + \psi_4 Z_{c(i),t-1} + \omega_{c(i)} + \tau_t + v_{ict}$ where $\widehat{WPEI}_{c,t-1}$ denotes the fitted values of women political empowerment from the first stage.

²⁰First-stage estimates differ slightly across outcomes because different attitude variables have different rates of missing data, resulting in varying sample sizes.

irrelevant.²¹ As a further robustness check, we compute Anderson-Rubin (1949) confidence intervals, which remain valid under weak instruments. These confirm the direction and significance of our TSLS estimates and are available upon request.

Table 9 presents the second stage results. The direction of the effects matches our earlier OLS and ordered logit estimates across all outcomes. However, the TSLS estimates are noticeably larger in magnitude than the corresponding OLS estimates, implying that the OLS estimates are downward biased.²² Another complementary explanation lies in what the TSLS actually identifies: a Local Average Treatment Effect (LATE), specific to compliers. The causal effects for this complier subpopulation may well exceed the effects for the entire population. Complier countries — those where gender quota adoption drove rapid increases in female political empowerment — are likely the countries where identity-signalling and agenda-setting effects on citizens’ attitudes are strongest. By contrast, countries with historically high female empowerment, such as some Nordic countries, where attitudes may already have adjusted over time, tend to exhibit smaller marginal effects. Because OLS averages across the full sample, including these latter cases, its estimates may be attenuated relative to those from TSLS.

The estimated effects are economically meaningful. Within the environmental concern and beliefs domain, the most pronounced effect is for ‘Future Environment vs. Current Jobs’, where a one-standard-deviation increase in the instrumented index raises the probability of expressing a pro-environmental attitude by 33 percentage points. In the willingness to pay or sacrifice domain, the largest effect is observed in citizens’ willingness to pay higher prices for environmental protection — with a 27-percentage-point decline in the corresponding probability. Within risk perception, the same increase raises the perceived danger of car pollution by 18 percentage points — the strongest effect in that domain. These patterns are intuitive: the outcomes most substantially affected are plausibly those most directly

²¹The F-statistics also exceed the critical value of the Stock-Yogo (2005) 10% maximal IV size in all cases.

²²Measurement error in expert-survey-based empowerment indices can attenuate OLS estimates downward (Coppedge et al., 2016), despite careful construction of the indices to avoid such errors.

tied to citizens' everyday economic trade-offs and immediate environmental exposures. The summary indices in Table 10 corroborate these findings across all three outcome categories.

To put the economic magnitude of our estimates into perspective, consider the following example. A country at the 75th percentile of women's political empowerment differs from one at the 25th percentile by 0.837 standard deviations. Our TSLS estimate implies that this gap alone corresponds to a 15 percentage-point difference in the probability of perceiving car pollution as a serious risk between the two countries.²³

7.2 Partial Identification

A natural concern with our instrumental variable strategy is that gender quota adoption might affect environmental attitudes through channels other than women's political empowerment, which would violate the exclusion restriction. While we cannot control for all possible confounders, we can assess how serious this concern is. We use the partial identification approach of Oster (2019) to examine how large the influence of unobservables would need to be, relative to observables, to overturn our estimated effects. A ratio well above 1 would suggest that unobservables are unlikely to drive our results. We compute the bias-adjusted coefficients using the specification below.

$$\beta_1^* \approx \tilde{\beta}_1 - \delta(\dot{\beta}_1 - \tilde{\beta}_1) \frac{R_{max} - \tilde{R}}{\tilde{R} - \dot{R}} \quad (3)$$

Here $\dot{\beta}_1$ and $\dot{R}$ denote the coefficient on women's political empowerment and the R-squared, respectively, from estimating specification (1) setting $\beta_2 = \beta_3 = 0$ (that is, excluding all individual and country-level controls and retaining only fixed effects). Similarly, $\tilde{\beta}_1$ and $\tilde{R}$ denote the corresponding estimates from the fully controlled specification where $\beta_2 \neq 0, \beta_3 \neq 0$.

Here δ captures the relative importance of unobservable variables compared to observable variables in explaining the treatment-outcome relationship, while R_{max} is the R-squared

²³The corresponding OLS estimate yields only a 6 percentage-point gap — consistent with the attenuation bias likely present in OLS.

that would be obtained if all relevant observable and unobservable variables were included in the regression. Both quantities are unknown. Oster(2019) proposes bounding the true causal effect between $\tilde{\beta}_1$ and β_1^* , where β_1^* is estimated for a given value of R_{max} under the assumption of $\delta = 1$ — denoting that observable and unobservable variables are equally important in explaining the outcome. We consider two values for R_{max} : first, $R_{max} = 0.6$, given the substantial idiosyncratic variation typical of individual-level attitude data; and second, $R_{max} = \min(1.3\tilde{R}, 1)$, following the rule of thumb proposed by Oster (2019). Our results are robust to alternative values of $R_{max} \in [0.4, 0.6]$.

Table 11 shows that the identified sets reported in columns (3) and (5) exclude zero for every outcome, indicating that the estimated effects of women’s political empowerment on pro-environmental attitudes are at least partially causal. These bounds combine the fully controlled estimate $\tilde{\beta}_1$ from specification (1) in Table 3 with the bias adjusted estimate β_1^* , calculated from equation (3) under the assumption of $\delta = 1$ for each of the two alternative values of R_{max} considered. Columns (2) and (4) report the values of $\hat{\delta}$ that would be required to drive the estimated treatment effect to zero under each assumption about R_{max} . In every case, $\hat{\delta}$ exceeds one, implying that selection on unobservables would have to be stronger than selection on observables to eliminate the estimated effects.²⁴ For a couple of cases, $\hat{\delta}$ exceeds 15 — far beyond this threshold — offering strong evidence that our estimated effects are robust to omitted variable bias.

7.3 Heterogeneity by Gender

In this section, we examine how women’s political empowerment affects the environmental attitudes of men and women separately. For the environmental concern, beliefs and risk perception variables, two competing mechanisms predict differential effects by gender. First, empowerment may work through a role model effect. Female politicians publicly

²⁴In several cases, $\hat{\delta}$ is negative, implying that unobservables would need to work in the opposite direction to observables to eliminate the treatment effect, which we have no clear reason to expect. This lends support to the robustness of our estimates.

champion environmental causes, validating the environmental values of women citizens and giving them greater confidence to express views they may have previously felt were marginalised. Under this mechanism, effects should be concentrated among women. Second, empowerment’s effects may also reflect identity-signalling and agenda-setting. Female politicians make environmental issues more salient in public discourse through speeches, policy, and media coverage — potentially shifting attitudes across society over time. This channel can affect both men and women, though the magnitudes may differ depending on their baseline levels of environmental concern. Which of these mechanisms is supported by the data is ultimately an empirical question.

Table 12 presents the TSLS results.²⁵ The instrumented women’s political empowerment index significantly improves pro-environmental attitudes for both men and women across all environmental concern and risk perception variables, consistent with our second mechanism. By contrast, it weakens such attitudes in the willingness to pay domain for both genders, which fits the political substitution effect discussed earlier. The confidence intervals for men and women overlap throughout, indicating no statistically detectable difference in effects by gender. When we construct Anderson indices for the three broad outcome categories, women’s empowerment raises pro-environmental attitudes for both men and women in the concern and risk perception domains, and lowers them in the willingness-to-act domain.

This coherence — across genders, outcome measures, treatment dimensions, and estimation methods — makes it unlikely that the results occur purely by chance. This suggests that women’s political empowerment plays a meaningful role in transforming citizens’ environmental preferences.

We also considered heterogeneity by development status. We do not report these results, as our sample contains only a small share of developing countries, leaving us underpowered to detect IV-TSLS effects within this subgroup; any findings would also not be generalisable to the broader developing world.

²⁵See Tables A3 and A4 for the corresponding OLS results.

7.4 Placebo Tests

A key concern with any instrumental variable strategy is whether the instrument is truly exogenous. To address this, we run a placebo test, restricting the sample to the pre-adoption period and testing whether *eventual* gender quota adoption predicts citizens' environmental attitudes before quotas are actually implemented. If our instrument is valid, citizens in countries that eventually adopt a quota should show no systematic difference in environmental attitudes from those in non-adopting countries during the pre-adoption period. Table 13 shows these results. The reduced-form coefficients on eventual quota adoption are small and statistically insignificant across all outcome variables, though estimated imprecisely in some cases due to limited statistical power. This falsification exercise supports the validity of our identification strategy.

7.5 Multiple Hypothesis Testing

As we test a large number of outcomes, a natural question arises about multiple inference: some coefficients could appear significant even absent any true effect. To address this, we adjust our p values to account for the fact that we are testing many outcomes at once. Specifically, we control the False Discovery Rate (FDR), defined as the expected share of rejected hypotheses that are actually false rejections (Type I errors). We rely on this approach rather than alternatives because it trades a small increase in Type I errors for a meaningful gain in statistical power. We control the FDR using the two stage method proposed by Benjamini, Krieger, and Yekutieli (2006), applying Anderson(2008)'s implementation. This procedure yields a q-value for each tested hypothesis, which is the multiple testing counterpart to the conventional p value. A test's q-value represents the lowest FDR threshold at which that test and every other test with an equal or smaller p-value — would still be classified as significant.

Table A5 shows that, across all thirteen outcomes, every result significant in our primary TSLS specification with controls remains significant after controlling the FDR at $q=0.05$. This lends additional confidence to our instrumental variable estimates.

7.6 Alternative Estimators

Given potential concerns about whether our gender quota instrument satisfies the exclusion restriction, we adopt a set of alternative identification strategies to check the robustness of our TSLS results. We begin with the two stage least squares estimator proposed by Lewbel (2012), known as Lewbel IV-TSLS. What sets this estimator apart is that it does not depend on the exclusion restriction required by conventional IV estimators. Instead, it deals with endogeneity by building its own instruments from within the model, exploiting patterns of heteroscedasticity in the data. These internal instruments are formed by taking the residuals from auxiliary regressions and interacting them with each exogenous regressor already included in the model, after centering those regressors around their means. The Lewbel approach allows us to use either only internally generated instruments or a combination of both internal and external instruments in a specification. This makes it especially valuable in settings where credible external instruments are unavailable or where researchers want to strengthen the precision of an IV estimator that already has an external instrument.

Table A6 shows the results from the Lewbel IV-TSLS estimator. These results are obtained using only internally generated instruments, and unsurprisingly, reliance on these internal instruments yields much larger standard errors than our main TSLS estimates. Reassuringly, however, the estimates are qualitatively similar to the baseline results across all individual attitude outcomes.

As a further robustness check on our TSLS results, we apply the Kinky Least Squares Estimator developed by Kiviet (2020). Rather than relying on an excluded instrument, this method places direct restrictions on how endogenous the regressor is allowed to be (Kripfganz and Kiviet, 2021). Kripfganz and Kiviet (2021) also show that, this approach can produce tighter confidence intervals than conventional two-stage least squares.

Table A7 presents these results. Consistent with our earlier checks, the point estimates closely track our primary TSLS specification.

Taken together, these results across multiple econometric techniques strengthen our central claim: exposure to women in power appears to shift citizens towards more pro-

environmental attitudes and greater demand for environmental protection. For policymakers, this shift matters — it points to a public more receptive to environmental measures, creating room to design policies that improve environmental outcomes.

8 Implications for Pro-Environmental behaviour

Having shown that women’s empowerment broadly shapes citizens’ pro-environmental attitudes, a natural next question is whether these attitudes translate into actual environmental behaviour. Specifically, does the heightened environmental concern and risk perception induced by women’s political empowerment lead people to act more sustainably? We examine this using the environmental behaviour variables available in the ISSP Environment Module. However, some of these variables are not consistently available across all four survey rounds, which makes our analysis underpowered to detect the underlying effects.

We consider seven pro-environmental behaviour variables drawn from the ISSP: avoiding environmentally harmful products, sorting recyclable waste, cutting back on car driving for environmental reasons, membership in an environmental group, signing an environmental petition, donating to an environmental group, and taking part in an environmental protest. Each variable is coded as a binary indicator, taking the value one if the respondent reports engaging in the behaviour and zero otherwise.

Table 14 presents the TSLS results for pro-environmental behaviour outcomes. The first stage Kleibergen-Paap F-statistic confirms that gender quota is not a weak instrument.²⁶ The instrumented women’s political empowerment significantly raises the probability that citizens sort recyclables, sign environmental petitions, and take part in environmental protests, by 28, 17, and 4 percentage points, respectively. It also increases the probability of avoiding environmentally harmful products and joining an environmental group, although

²⁶First stage coefficients and F-statistics vary marginally across outcomes due to missing observations and varying estimation sample sizes. The reported first stage corresponds to ‘Participated in environmental protest’. All first stage coefficients are positive and significant, with F-statistics exceeding the rule-of-thumb threshold of 10.

these estimates are imprecise. By contrast, women’s empowerment lowers the probability of donating money to an environmental group – a pattern consistent with the political substitution effect observed earlier for willingness-to-pay outcomes, since both donating and paying higher prices involve a direct personal financial cost. Women’s empowerment similarly reduces the probability that citizens cut back on car driving. This may reflect the fact that habits as entrenched as daily driving are slow to change, especially when people doubt that acting alone can meaningfully improve environmental outcomes without a wider shift in collective behaviour.

Overall, women’s political empowerment is linked to stronger pro-environmental behaviour across several dimensions, which is encouraging. That said, these results warrant some caution: behavioural self-reports may be more prone to social desirability bias than attitudinal responses. Even so, the findings suggest that women’s empowerment may do more than shift attitudes – it may also help narrow the attitude-behaviour gap, with potential downstream effects on environmental outcomes.

9 Mediation Analysis

Our findings point to a nuanced relationship between women’s political empowerment, environmental attitudes, and behaviour. Importantly, these attitudinal shifts are not mere cheap talk – they translate into real behavioural change among citizens.

What mechanisms drive these effects? In Section 3, we discuss three plausible channels: identity signalling effect, agenda setting effect and political substitution effect. Formally testing the first two mechanisms is beyond the scope of this paper, given the data limitations and identification challenges inherent in cross-country analysis. The third can be indirectly tested through observable proxies related to policies and institutions, as we do below.

The third lends itself to an indirect empirical test: we examine whether it operates through observable changes in institutional quality and environmental policy stringency, below. The existing literature identifies stricter environmental policy and stronger institutional quality as channels mediating the relationship between female empowerment and environmental

outcomes such as air quality (Rios et al., 2024). Our primary objective is to examine whether environmental policy and institutional quality serve as plausible pathways explaining our main effects on citizens’ environmental attitudes. For this purpose, we turn to suggestive causal mediation analysis (Imai et al., 2011; Carpena and Zia, 2020) that disentangles the share of the effect operating specifically through the hypothesized mediator and the share operating through the remaining channels. We estimate the following two specifications:

$$Mediator_{ict} = \alpha_0 + \beta_1 WPEI_{c,t-1} + \beta_2 X_{ict} + \beta_3 Z_{c(i),t-1} + \gamma_{c(i)} + \delta_t + \epsilon_{ict} \quad (4)$$

$$Y_{ict} = \psi_0 + \delta_1 WPEI_{c,t-1} + \delta_2 X_{ict} + \delta_3 Z_{c(i),t-1} + \delta_4 Mediator_{ict} + \pi_{c(i)} + \phi_t + v_{ict} \quad (5)$$

The mediation effect corresponds to the product of coefficients $\hat{\beta}_1 \hat{\delta}_4$, estimated via OLS from equations 4 and 5. The direct effect is captured by $\hat{\delta}_1$ in equation 5. Standard errors for the mediation effect are computed using non-parametric bootstrapping.

Due to space constraints, we report only the mediation effect estimates in Table 15. We consider three mediators – Environmental Policy Stringency, Government Effectiveness and Control of Corruption. Consistent with earlier evidence (Mavisakalyan and Tarverdi, 2019a; Swamy et al., 2001), our estimates from equation 4 show that women’s empowerment leads to stricter environmental policy and stronger institutional quality. However, none of the three mediators significantly explains the effect of women’s empowerment on citizens’ environmental concern and risk perception, as shown for the first nine outcome variables in Table 15. ²⁷The estimates are not only statistically insignificant, but also close to zero in magnitude. By contrast, the mediation effects are negative and statistically significant for the willingness-to-act variables in the last four rows of the table. This suggests that environmental policy and institutional quality are potential channels behind the negative effects of women’s political empowerment on citizens’ willingness to act. This is consistent with citizens delegating environmental responsibility to institutions rather than bearing the financial costs themselves – the mechanism we discuss in Section 3.

²⁷Other measures of institutional quality, obtained from the World Bank, yield similar results to those in Table 15.

Unlike objective outcomes like air quality, which women politicians can improve directly through policy, our null mediation results are informative in themselves. They suggest that our effects may operate through a different channel — the visibility and political salience of powerful women and the environmental issues they advocate for — rather than through the policy consequences of their empowerment. For policy stringency to shift citizens’ beliefs, a citizen must first become aware that policy has changed, correctly attribute that change to women leaders, and interpret it as informative about the problem’s severity. This is a long chain, and each step is unlikely to hold in sequence. The chain is further weakened by citizens’ generally limited or inaccurate knowledge of specific regulatory changes across environmental domains, making it improbable that such changes meaningfully update their beliefs. A second plausible reason is that our treatment variable — a broader, multidimensional measure of women’s political empowerment — captures more than descriptive political representation. As a result, its effects need not operate through enacted policy and the quality of its enforcement.

Our mediation results make it difficult to attribute the observed effects on concern and risk perception to the two most obvious literature-established channels - policy stringency and institutional quality. The remaining evidence is broadly consistent with the identity signalling and agenda setting mechanisms discussed above. A caveat is warranted here. We cannot establish these mechanisms directly with our data, and we do not claim to have pinned them down with the same precision an experimental design would allow. Any alternative story would need to account for the same pattern we find — attitudes shifting with no corresponding movement in either channel. Identity-signalling and agenda-setting mechanisms discussed in section 3 fit this pattern, since neither works through the implementation and enforcement of policy. We cannot rule out other unobserved belief based channels with the same property, but we are not aware of an equally plausible candidate in the existing literature.

10 Conclusion

Addressing climate change requires progress on two fronts: stricter environmental policy on the supply side, and citizens who value environmental preservation on the demand side. Economists have studied the supply side, but the demand side — what shapes citizens’ underlying environmental preferences in the first place — has received far less attention. This paper asks whether one overlooked factor, women’s political empowerment, contributes to that demand, and what the mechanisms behind it are.

To answer this, we combine rich individual-level data on environmental attitudes, behaviours, and sociodemographic characteristics from the ISSP with comprehensive country-level measures of economic development, structural characteristics, and political and institutional quality spanning more than two decades. Rather than relying on a narrow measure of political representation, we use a multidimensional index of women’s political empowerment that also captures their agency in decision making in various spheres of life. We first document a new association between female empowerment and environmental attitudes, then go further, using gender quota adoption as an instrument to establish that this relationship is plausibly causal.

The results tell a consistent story. Women’s political empowerment increases citizens’ environmental concern and heightens their sensitivity to environmental risk. We rule out institutional quality and regulatory stringency as explanations, and point instead to identity-signalling and agenda-setting effects arising from citizens’ exposure to women in power. Yet greater concern does not translate into greater personal financial commitment to safeguarding the environment. If anything, citizens appear to delegate environmental responsibility to empowered female leaders rather than incur these costs themselves. At the same time, these attitudinal shifts are not merely symbolic — they are accompanied by corresponding changes in behaviour, from recycling to environmental activism. These findings hold up across a wide range of checks — alternative samples, specifications, estimation methods, falsification exercises, and measures of both empowerment and attitudes.

Our sample is limited to a set of advanced and emerging market economies, since the

ISSP — despite offering rich cross-national coverage of environmental attitudes over time — includes few low-income countries. Our findings should therefore be interpreted with caution when considering their applicability to the broader developing world.

Nonetheless, these results carry important policy implications. Expanding women’s political voice may deliver environmental benefits that go well beyond the policies women in office directly enact. Whether through formal channels like quotas and reserved seats, or through the broader strengthening of women’s agency in civic and political life, these efforts can promote a lasting, citizen-level engagement on environmental issues. There may also be an additional benefit — as women’s political leadership strengthens institutions and governance, it can help build the public trust those institutions need to actually deliver on environmental promises.

Future research could build on our findings by testing the proposed mechanisms more directly. Data on media coverage of environmental issues and parliamentary debates, for instance, could offer more direct evidence for the agenda-setting channel we identify indirectly here. Given our data structure, we cannot observe how the same individual’s attitudes evolve with prolonged exposure to women’s empowerment. Panel data tracking the same individuals over time would offer a more direct test of such long-run effects, holding respondent composition fixed in a way a repeated cross-section cannot.²⁸ Beyond gender, other dimensions of politicians’ identity — such as age, education, or socio-economic background — may also determine citizens’ environmental attitudes, and warrant further investigation.

AI Disclosure

During the preparation of this work, the author used ChatGPT by OpenAI for language refinement and copy-editing. The author reviewed and edited the output as needed and take full responsibility for the content of the published article.

²⁸Note that, our results still hold when restricting the sample to earlier survey waves (1993-2010), suggesting they are not driven by later periods alone.

Table 1: Construction of Binary Pro-environmental Attitude Variables

Variable	Survey Question (Brief)	Pro- environmental Response	Description (Brief)
Future Environment vs. Current Jobs	Worry too much about environment and not enough about prices and jobs	Strongly disagree, Disagree	Disagreeing reflects prioritisation of long-term environmental protection over short-term economic concerns.
Modern Life Harms Environment	Almost everything we do in modern life harms the environment	Strongly agree, Agree	Agreeing indicates awareness of human impact on nature and the environmental consequences of everyday economic activity.
Progress Harms Environment	Worry too much about progress harming environment	Strongly disagree, Disagree	Disagreeing signals scepticism towards growth-centric narratives that downplay environmental damage.
Personal Environmental Agency	Too difficult to do much about environment	Strongly disagree, Disagree	Disagreeing reflects a sense of individual efficacy and responsibility towards environmental protection.
Willingness to Pay Higher Prices	Protect environment: pay much higher prices	Very willing, Fairly willing	Reflects readiness to bear private financial costs for collective environmental benefits.

Table 1 continued

Variable	Survey Question (Brief)	Pro- environmental Response	Description (Brief)
Willingness to Pay Higher Taxes	Protect environment: pay much higher taxes	Very willing, Fairly willing	Signals support for public policy interventions aimed at environmental protection and acceptance of redistributive mechanisms to finance such efforts..
Willingness to Accept Lower Living Standards	Protect environment: cut your standard of living	Very willing, Fairly willing	Reflects strong commitment to environmental goals beyond symbolic concern to material sacrifice.
Costly Environmental Action	I do what is right, even when it costs more money and time	Strongly agree, Agree	Reflects normative commitment to environmental protection independent of personal economic incentives.
Risk of Car Pollution	Air pollution caused by cars is (dangerous/not dangerous) for environment	Extremely dangerous, Very dangerous	Reflects concern about anthropogenic environmental risks and their health consequences.
Risk of Industrial Air Pollution	Air pollution caused by industry is (dangerous/ not dangerous) for environment	Extremely dangerous, Very dangerous	Captures recognition of industrial activity as a major source of environmental harm.

Table 1 continued

Variable	Survey Question (Brief)	Pro- environmental Response	Description (Brief)
Risk of Agricultural Chemicals	Pesticides used in farming are (dangerous/ not dangerous) for environment	Extremely dangerous, Very dangerous	Reflects concern about environmental contamination and ecological degradation from modern agricultural practices.
Risk of Water Pollution	Pollution of rivers is (dangerous/ not dangerous) for environment	Extremely dangerous, Very dangerous	Indicates awareness of environmental damage and its ecological and social consequences.
Risk of Global Temperature Rise	A rise in world's temperature is (dangerous/ not dangerous) for environment	Extremely dangerous, Very dangerous	Captures concern about one of the central global environmental challenges.

Table 2: Summary Statistics

Variable	Source	Mean	Std. Dev.
Log GDP Per Capita*	World Bank, UN WPP(2024)	10.513	0.581
Log Trade Openness	World Bank	4.178	0.462
Services Employment (%)	World Bank/ILO	67.449	8.919
Industry Employment (%)	World Bank/ILO	25.372	6.154
Agricultural Employment (%)	World Bank/ILO	7.179	7.232
EPS Index	OECD	1.820	2.067
Oil+gas NX per capita (real USD)#	Ross & Mahdavi(2015)	-4.147	4.743
Control of Corruption^	World Bank WGI	79.458	22.503
Government Effectiveness^	World Bank WGI	82.092	17.252
Polity Score	Polity V	8.904	3.098
Log Population Density	World Bank	4.122	1.387
Doing Business Rank	World Bank	37.653	32.528
Chief Executive Party: Left (Share)	DPI (IADB)	0.417	0.493
Chief Executive Party: Right (Share)	DPI (IADB)	0.331	0.470
Female	ISSP	0.524	0.499
Age	ISSP	46.268	17.490
Years of schooling	ISSP	13.468	15.089
Share of individuals in urban areas	ISSP	0.693	0.461
N	138,610		

* PPP (constant 2021 international dollars).

IHS transformation is used to accommodate zero and negative values

^ Score : 0-100.

Table 3: OLS Estimates- Individual Attitude Variables

Outcome	Without Controls	With controls
Future Environment vs Current Jobs	0.0053 (0.0590)	0.0462** (0.0184)
Modern Life Harms Environment	0.0150 (0.0270)	0.0424** (0.0184)
Progress Harms Environment	0.0150 (0.0567)	0.0681** (0.0288)
Personal Environmental Agency	0.0322 (0.0455)	0.0638*** (0.0171)
Willingness to Pay Higher Prices	-0.0391 (0.0377)	-0.1156*** (0.0198)
Willingness to Pay Higher Taxes	-0.0381 (0.0401)	-0.1085*** (0.0196)
Willingness to Accept Lower Living Standards	-0.0328 (0.0297)	-0.1003*** (0.0224)
Costly Environmental Action	-0.0047 (0.0407)	-0.1527*** (0.0286)
Risk of Car Pollution	0.0469 (0.0589)	0.0663*** (0.0218)
Risk of Industrial Air Pollution	0.1171* (0.0625)	0.0888*** (0.0246)
Risk of Agricultural Chemicals	0.0797** (0.0336)	0.1763*** (0.0271)
Risk of Water Pollution	0.0476 (0.0400)	0.0570** (0.0263)
Risk of Global Temperature Rise	0.1305** (0.0596)	0.1233*** (0.0418)
N	138,610	138,610
Adjusted R^2	0.094	0.370

Notes: Column (1) lists the binary outcome variables. Column (2) presents estimates without controls; Column (3) includes the full set of controls. Each cell reports the coefficient on the standardised Women's Political Empowerment Index from a separate regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: OLS Estimates- Summary Indices

Outcome	Index: (Anderson, 2008)	Index: (Anderson, 2008)	Index: PCA	Index: PCA	Index: Simple Average	Index: Simple Average
Environmental Concern	0.0178 (0.0888)	0.2160*** (0.0492)	0.0276 (0.1490)	0.3075*** (0.0727)	0.0270 (0.0819)	0.1362*** (0.0358)
Willingness to Pay	-0.1038 (0.1021)	-0.1708*** (0.0404)	-0.2172 (0.1711)	-0.3305*** (0.0611)	-0.1122 (0.1064)	-0.2053*** (0.0418)
Environmental Risk Perception	0.3105** (0.1252)	0.1661*** (0.0431)	0.5430** (0.2244)	0.3132*** (0.0703)	0.2149** (0.0859)	0.1122*** (0.0293)
N	138,610	138,610	138,610	138,610	138,610	138,610
Adjusted R^2	0.088	0.388	0.088	0.394	0.088	0.425

Notes: Column (1) lists the outcome domain. Columns (2), (4) and (6) present estimates without controls; Columns (3), (5) and (7) includes the full set of controls. Each row presents estimates for a different outcome domain, using three alternative aggregation methods — Anderson (2008), Principal Component Analysis, and simple averaging. The Environmental Concern Index aggregates three variables: Future Environment vs. Current Jobs, Modern Life Harms Environment, and Progress Harms Environment. The Willingness to Pay Index aggregates three variables: Willingness to Pay Higher Prices, Willingness to Pay Higher Taxes, and Willingness to Accept Lower Living Standards. The Environmental Risk Perception Index aggregates five variables: Risk of Car Pollution, Risk of Industrial Air Pollution, Risk of Agricultural Chemicals, Risk of Water Pollution, and Risk of Global Temperature Rise. Each cell reports the coefficient on the standardised Women’s Political Empowerment Index from a separate regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: OLS Estimates- Components of the Women Political Empowerment Index

Outcome	WPPI	WCSPi	WCLI
Future Environment vs Current Jobs	0.1162*** (0.0327)	0.1733*** (0.0171)	0.0071 (0.0189)
Modern Life Harms Environment	0.1266*** (0.0400)	0.0433** (0.0194)	0.0066 (0.0318)
Progress Harms Environment	0.0814** (0.0339)	0.0768*** (0.0174)	0.1079*** (0.0245)
Personal Environmental Agency	0.0533** (0.0211)	0.0345*** (0.0083)	0.0456** (0.0177)
Willingness to Pay Higher Prices	-0.1048*** (0.0196)	0.0035 (0.0104)	-0.1645*** (0.0207)
Willingness to Pay Higher Taxes	-0.0823*** (0.0205)	0.0092 (0.0133)	-0.1640*** (0.0231)
Willingness to Accept Lower Living Standards	-0.0841*** (0.0240)	0.0202 (0.0256)	-0.0782*** (0.0280)
Costly Environmental Action	-0.4611*** (0.0736)	-0.0169 (0.0263)	-0.0813*** (0.0277)
Risk of Car Pollution	0.1062*** (0.0358)	0.1254*** (0.0172)	0.0496** (0.0229)
Risk of Industrial Air Pollution	0.0926*** (0.0237)	0.0946*** (0.0187)	0.0686*** (0.0245)
Risk of Agricultural Chemicals	0.1526*** (0.0661)	0.0856*** (0.0196)	0.0690*** (0.0246)
Risk of Water Pollution	0.1631*** (0.0376)	0.0376** (0.0182)	0.0111 (0.0204)
Risk of Global Temperature Rise	0.1324*** (0.0440)	0.1773*** (0.0270)	0.0207 (0.0197)
N	138,610	138,610	138,610
Adjusted R^2	0.332	0.366	0.391

Notes: Column (1) lists the outcome variables, coded as binary. Column (2) presents estimates using the standardised Women's Political Participation Index as the treatment variable; Column (3) presents estimates using the standardised Women's Civil Society Participation Index; Column (4) presents estimates using the standardised Women's Civil Liberties Index. Each cell reports the coefficient from a separate regression. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: TSLS Estimates Controlling for Postmaterialist Share

Outcome	Without Control	With Control
Future Environment vs Current Jobs	0.3339*** (0.0369)	0.2858*** (0.0347)
Modern Life Harms Environment	0.2754*** (0.1020)	0.2697*** (0.0940)
Progress Harms Environment	0.1921*** (0.0446)	0.1500*** (0.0347)
Personal Environmental Agency	0.1897*** (0.0219)	0.2497*** (0.0216)
Willingness to Pay Higher Prices	-0.2746** (0.1195)	-0.2520*** (0.0738)
Willingness to Pay Higher Taxes	-0.1317*** (0.0442)	-0.1027*** (0.0288)
Willingness to Accept Lower Living Standards	-0.1601** (0.0728)	-0.1178*** (0.0542)
Costly Environmental Action	-0.2303*** (0.0867)	-0.2178** (0.0864)
Risk of Car Pollution	0.1804*** (0.0491)	0.1726*** (0.0411)
Risk of Industrial Air Pollution	0.1524*** (0.0506)	0.1524*** (0.0522)
Risk of Agricultural Chemicals	0.1761*** (0.0481)	0.1839*** (0.0306)
Risk of Water Pollution	0.1641*** (0.0544)	0.1656*** (0.0452)
Risk of Global Temperature Rise	0.1719** (0.0720)	0.1621** (0.0643)
N	138,610	138,610
Adjusted R^2	0.318	0.350
First stage coefficient	0.4201*** (0.0810)	0.3014*** (0.0614)
Kleibergen-Paap F statistic	26.91	24.09

Notes: Column (1) lists the binary outcome variables. Column (2) presents estimates without postmaterialist share as a control from Table 9; Column (3) presents the estimates with postmaterialist share as a control. Each cell reports the second-stage coefficient on the standardised Women's Political Empowerment Index (instrumented by gender quota adoption) from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level are in parentheses. First-stage coefficients and F-statistics are also provided. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: OLS Estimates- Exclusion Restriction of Gender Quota Instrument

Control variables		
	(2)	(3)
Panel A: Future Environment vs Current Jobs		
WPEI		0.1320*** (0.0388)
Gender quota	0.1296*** (0.0290)	-0.0062 (0.0284)
Panel B: Modern Life Harms Environment		
WPEI		0.0966*** (0.0166)
Gender quota	0.0931*** (0.0128)	0.0104 (0.0356)
Panel C: Progress Harms Environment		
WPEI		0.0839*** (0.0041)
Gender quota	0.0759*** (0.0084)	-0.0201 (0.0190)
Panel D: Personal Environmental Agency		
WPEI		0.0750*** (0.0222)
Gender quota	0.0746*** (0.0174)	-0.0012 (0.0126)
Panel E: Willingness to Pay Higher Prices		
WPEI		-0.1063*** (0.0361)
Gender quota	-0.1052*** (0.0366)	-0.0246 (0.0494)
Panel F: Willingness to Pay Higher Taxes		
WPEI		-0.0762*** (0.0151)
Gender quota	-0.0558*** (0.0130)	-0.0309 (0.0299)
Panel G: Willingness to Accept Lower Living Standards		
WPEI		-0.0464*** (0.0103)
Gender quota	-0.0653*** (0.0247)	-0.0014 (0.0358)
Panel H: Costly Environmental Action		
WPEI		-0.0699** (0.0295)
Gender quota	-0.0417*** (0.0116)	0.0356 (0.0394)
Panel I: Risk of Car Pollution		
WPEI		0.0433*** (0.0184)
Gender quota	0.0683*** (0.0186)	0.0022 (0.0096)

Continued on next page

Table 7 – continued

Control variables		
	(2)	(3)
Panel J: Risk of Industrial Air Pollution		
WPEI		0.0479*** (0.0155)
Gender quota	0.0230*** (0.0099)	0.0125 (0.0107)
Panel K: Risk of Agricultural Chemicals		
WPEI		0.0865** (0.0360)
Gender quota	0.0313*** (0.0085)	0.0308 (0.0243)
Panel L: Risk of Water Pollution		
WPEI		0.0474** (0.0184)
Gender quota	0.0881*** (0.0328)	0.0428 (0.0491)
Panel M: Risk of Global Temperature Rise		
WPEI		0.0879** (0.0399)
Gender quota	0.0789*** (0.0356)	0.0427 (0.0362)
N	138,610	138,610
Adjusted R^2	0.341	0.341

Notes: Column (2) reports the OLS coefficient on gender quota from a regression of the outcome on gender quota alone. Column (3) reports the coefficients on gender quota and WPEI from a regression of the outcome on both these variables. Each panel corresponds to a separate outcome variable. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: First Stage Estimates- Gender Quota as Instrument (Individual Attitude Variables)

Outcome	Estimates
Future Environment vs Current Jobs	0.4201*** (0.0810) KPF- 26.91
Modern Life Harms Environment	0.4190*** (0.0809) KPF- 26.84
Progress Harms Environment	0.4234*** (0.0823) KPF- 26.47
Personal Environmental Agency	0.4203*** (0.0802) KPF- 27.48
Willingness to Pay Higher Prices	0.4204*** (0.0807) KPF- 27.15
Willingness to Pay Higher Taxes	0.4237*** (0.0814) KPF- 27.09
Willingness to Accept Lower Living Standards	0.4197*** (0.0807) KPF- 27.06
Costly Environmental Action	0.4225*** (0.0806) KPF- 27.47
Risk of Car Pollution	0.4176*** (0.0790) KPF- 27.9
Risk of Industrial Air Pollution	0.4203*** (0.0794) KPF- 27.99
Risk of Agricultural Chemicals	0.4173*** (0.0795) KPF- 27.54
Risk of Water Pollution	0.4183*** (0.0793) KPF- 27.81
Risk of Global Temperature Rise	0.4222*** (0.0835) KPF-25.54

Notes: Column (1) lists the binary outcome variables. Column (2) presents the first stage estimates and the Kleibergen-Paap F-statistics. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: TSLS Estimates- Gender Quota as Instrument (Individual Attitude Variables)

Outcome	(Without controls)	(With controls)
Future Environment vs Current Jobs	0.1083 (0.1266)	0.3339*** (0.0369)
Modern Life Harms Environment	0.2403* (0.1271)	0.2754*** (0.1020)
Progress Harms Environment	0.1545 (0.1566)	0.1921*** (0.0446)
Personal Environmental Agency	0.0103 (0.1160)	0.1897*** (0.0219)
Willingness to Pay Higher Prices	-0.1465 (0.1541)	-0.2746*** (0.1195)
Willingness to Pay Higher Taxes	-0.0454 (0.0969)	-0.1317*** (0.0442)
Willingness to Accept Lower Living Standards	-0.0190 (0.2042)	-0.1601** (0.0728)
Costly Environmental Action	-0.1240 (0.0999)	-0.2303*** (0.0867)
Risk of Car Pollution	0.0649 (0.0566)	0.1804*** (0.0491)
Risk of Industrial Air Pollution	0.1754 (0.1940)	0.1524*** (0.0506)
Risk of Agricultural Chemicals	0.3466 (0.3095)	0.1761*** (0.0481)
Risk of Water Pollution	0.2417 (0.3295)	0.1641*** (0.0544)
Risk of Global Temperature Rise	0.2596 (0.3224)	0.1719** (0.0720)
N	138,610	138,610
Adjusted R^2	0.075	0.318

Notes: Column (1) lists the binary outcome variables. Column (2) presents estimates without controls; Column (3) includes the full set of controls. Each cell reports the second-stage coefficient on the standardised Women's Political Empowerment Index (instrumented by gender quota adoption) from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: TSLS Estimates- Gender Quota as Instrument (Summary Indices)

Outcome	Index: Anderson (2008)	Index: Anderson (2008)	Index: PCA	Index: PCA	Index: Simple Average	Index: Simple Average
Environmental Concern	0.3602 (0.2296)	0.2039** (0.0915)	0.1149 (0.2603)	0.2232** (0.0930)	0.1200 (0.2202)	0.2066*** (0.0744)
Willingness to Pay	-0.3298 (0.2780)	-0.2021** (0.0963)	-0.2982 (0.1908)	-0.2838** (0.1402)	-0.1986 (0.1256)	-0.2145** (0.0954)
Environmental Risk Perception	0.2790* (0.1445)	0.3132*** (0.0788)	0.4290* (0.2428)	0.4407*** (0.1396)	0.1902* (0.1004)	0.2310*** (0.0778)
N	138,610	138,610	138,610	138,610	138,610	138,610
Adjusted R^2	0.075	0.307	0.076	0.304	0.073	0.348

Notes: Column (1) lists the outcome domain. Columns (2), (4) and (6) present estimates without controls; Columns (3), (5) and (7) includes the full set of controls. Each row presents estimates for a different outcome domain, using three alternative aggregation methods — Anderson (2008), Principal Component Analysis, and simple averaging. The Environmental Concern Index aggregates three variables: Future Environment vs. Current Jobs, Modern Life Harms Environment, and Progress Harms Environment. The Willingness to Pay Index aggregates three variables: Willingness to Pay Higher Prices, Willingness to Pay Higher Taxes, and Willingness to Accept Lower Living Standards. The Environmental Risk Perception Index aggregates five variables: Risk of Car Pollution, Risk of Industrial Air Pollution, Risk of Agricultural Chemicals, Risk of Water Pollution, and Risk of Global Temperature Rise. Each cell reports the second-stage coefficient on the standardised Women's Political Empowerment Index (instrumented by gender quota adoption) from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Test of Omitted Variable Bias- Oster test

Outcome	$ \hat{\delta} _{R_{max}=0.6}$	$[\tilde{\beta}_1, \beta_1^*]_{R_{max}=0.6, \delta=1}$	$ \hat{\delta} _{R_{max}=\min\{1.3\tilde{R}, 1\}}$	$[\tilde{\beta}_1, \beta_1^*]_{R_{max}=\min(1.3\tilde{R}, 1), \delta=1}$
Future Environment vs Current Jobs	1.714	[0.0462, 0.0732]	3.020	[0.0462, 0.0615]
Modern Life Harms Environment	2.215	[0.0424, 0.0615]	4.179	[0.0424, 0.0526]
Progress Harms Environment	1.366	[0.0681, 0.1180]	3.395	[0.0681, 0.0882]
Personal Environmental Agency	2.730	[0.0638, 0.0872]	5.104	[0.0638, 0.0763]
Willingness to Pay Higher Prices	1.712	[-0.1831, -0.1156]	4.006	[-0.1445, -0.1156]
Willingness to Pay Higher Taxes	2.190	[-0.1580, -0.1085]	4.115	[-0.1349, -0.1085]
Willingness to Accept Lower Living Standards	1.665	[-0.1606, -0.1003]	3.947	[-0.1257, -0.1003]
Costly Environmental Action	1.085	[-0.2935, -0.1527]	2.649	[-0.2104, -0.1527]
Risk of Car Pollution	4.142	[0.0663, 0.0823]	8.966	[0.0663, 0.0737]
Risk of Industrial Air Pollution	3.747	[0.0651, 0.0888]	8.149	[0.0779, 0.0888]
Risk of Agricultural Chemicals	2.168	[0.1763, 0.2576]	4.818	[0.1763, 0.2129]
Risk of Water Pollution	8.769	[0.0570, 0.0635]	16.106	[0.0570, 0.0605]
Risk of Global Temperature Rise	2.644	[0.0767, 0.1233]	25.687	[0.1185, 0.1233]

Notes: Column (1) lists the outcome variables. Columns (2) and (4) report $|\hat{\delta}|$ — the degree of selection on unobservables relative to observables required to eliminate the estimated effect of women’s political empowerment — computed assuming $R_{max} = 0.6$ and $R_{max} = \min(1.3\tilde{R}, 1)$ respectively. Columns (3) and (5) report the bounds on the treatment effect β_1 , computed under the assumption that unobservables are as important as observables in explaining the outcome (i.e. $\delta = 1$) for the same two values of R_{max} . See the main text for definitions of $\tilde{\beta}_1$ and $\tilde{R}$.

Table 12: TSLS Estimates- Heterogeneity Effects by Gender (Individual Attitude Variables)

Outcome	Male	Female
Future Environment vs Current Jobs	0.3131*** (0.0309)	0.3494*** (0.0423)
Modern Life Harms Environment	0.2686*** (0.0937)	0.2956** (0.1188)
Progress Harms Environment	0.2009*** (0.0573)	0.1858*** (0.0581)
Personal Environmental Agency	0.2743*** (0.0296)	0.1287*** (0.0276)
Willingness to Pay Higher Prices	-0.2243** (0.1099)	-0.3175** (0.1284)
Willingness to Pay Higher Taxes	-0.1356** (0.0603)	-0.1236*** (0.0330)
Willingness to Accept Lower Living Standards	-0.1419** (0.0640)	-0.1891*** (0.0659)
Costly Environmental Action	-0.2886*** (0.1079)	-0.1796** (0.0723)
Risk of Car Pollution	0.2125*** (0.0742)	0.1577** (0.0645)
Risk of Industrial Air Pollution	0.1546** (0.0638)	0.1546*** (0.0425)
Risk of Agricultural Chemicals	0.1462** (0.0684)	0.2044*** (0.0667)
Risk of Water Pollution	0.1860** (0.0765)	0.1071** (0.0499)
Risk of Global Temperature Rise	0.1298* (0.0679)	0.2140*** (0.0785)
N	63,496	74,897
Adjusted R^2	0.361	0.335
First stage coefficient	0.3850*** (0.0943)	0.3905*** (0.0916)
Kleibergen-Paap F statistic	16.66	18.16

Notes: Column (1) lists the binary outcome variables. Column (2) restricts the sample to male individuals; Column (3) restricts the sample to female individuals. Each cell reports the second-stage coefficient on the standardised Women's Political Empowerment Index (instrumented by gender quota adoption) from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. First-stage coefficients and F-statistics are also provided. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: Placebo Test- Pre-Adoption Individual Environmental Attitudes of Eventual Quota-Adopting Countries

Outcome	(With controls)
Future Environment vs Current Jobs	-0.0095 (0.0506)
Modern Life Harms Environment	0.0204 (0.0482)
Progress Harms Environment	0.0018 (0.0149)
Personal Environmental Agency	-0.0076 (0.0142)
Willingness to Pay Higher Prices	-0.0082 (0.0133)
Willingness to Pay Higher Taxes	-0.0061 (0.0649)
Willingness to Accept Lower Living Standards	0.0162 (0.0137)
Costly Environmental Action	0.0297 (0.0643)
Risk of Car Pollution	0.0590 (0.0490)
Risk of Industrial Air Pollution	0.0223 (0.0364)
Risk of Agricultural Chemicals	0.0691 (0.0593)
Risk of Water Pollution	0.0289 (0.0515)
Risk of Global Temperature Rise	0.0389 (0.0479)
N	74,310
Adjusted R^2	0.347

Notes: Column (1) lists the binary outcome variables. Each cell reports the reduced form coefficient on eventual quota-adopting countries in the pre-adoption period from a separate regression. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: TSLS Estimates- Gender Quota as Instrument(Individual behaviour Variables)

Outcome	With Controls
Avoid environmentally harmful products	0.1686 (0.1388)
Sort recyclables for environment	0.2791** (0.1340)
Cut back car driving for env. reasons	-0.1839*** (0.0702)
Member of environmental group	0.0429 (0.0306)
Signed a petition	0.1750** (0.0806)
Given money to environmental group	-0.0822*** (0.0217)
Participated in environmental protest	0.0398** (0.0165)
N	138,610
Adjusted R^2	0.254
First stage coefficient	0.4315*** (0.1154)
Kleibergen-Paap F statistic	13.97

Notes: Column (1) lists the binary outcome variables. Column (2) presents estimates with the full set of individual and country level controls. Each cell reports the second-stage coefficient on the standardised Women's Political Empowerment Index (instrumented by gender quota adoption) from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. First-stage coefficients and F-statistics are also provided. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15: Mediation Analysis Exploring Channels Through Which Women's Political Empowerment Affects People's Pro-Environmental Attitudes

Outcome	Mediation Effect		
	Environmental Policy Stringency	Government Effectiveness	Control of Corruption
Future Environment vs Current Jobs	0.0035 (0.0024)	0.0002 (0.0039)	0.0138 (0.0115)
Modern Life Harms Environment	0.0010 (0.0030)	-0.0041 (0.0031)	0.0032 (0.0101)
Progress Harms Environment	0.0016 (0.0015)	0.0004 (0.0029)	0.0055 (0.0090)
Personal Environmental Agency	0.0018 (0.0016)	-0.0024 (0.0031)	0.0064 (0.0144)
Risk of Car Pollution	0.0007 (0.0016)	-0.0057 (0.0038)	-0.0070 (0.0108)
Risk of Industrial Air Pollution	0.0010 (0.0014)	-0.0005 (0.0022)	0.0125 (0.0083)
Risk of Agricultural Chemicals	0.0023 (0.0019)	-0.0033 (0.0033)	0.0171 (0.0120)
Risk of Water Pollution	0.0027 (0.0018)	-0.0014 (0.0012)	0.0130 (0.0085)
Risk of Global Temperature Rise	0.0011 (0.0012)	-0.0028 (0.0027)	-0.0115 (0.0123)
Willingness to Pay Higher Prices	-0.0021** (0.0009)	-0.0020* (0.0012)	-0.0113*** (0.0039)
Willingness to Pay Higher Taxes	-0.0015* (0.0008)	-0.0057*** (0.0011)	-0.0139* (0.0073)
Willingness to Accept Lower Living Standards	-0.0020** (0.0009)	-0.0041*** (0.0012)	-0.0109*** (0.0039)
Costly Environmental Action	-0.0019** (0.0009)	-0.0027** (0.0012)	-0.0137*** (0.0044)

Notes: Column (1) lists the outcome variables. Columns (2)-(4) present the mediation effects, represented by the product of the coefficients $\hat{\beta}_1\hat{\delta}_4$ as described in the text, corresponding to the mediators -- Environmental Policy Stringency, Government Effectiveness, and Control of Corruption, respectively. All specifications include control variables, country and year fixed effects, and are weighted using ISSP survey weights. Bootstrapped standard errors, clustered at the country level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Gendered Governance and Environmental Attitudes: Does Political Agency Matter?

Supplementary Appendix

September 30, 2026

A Appendix

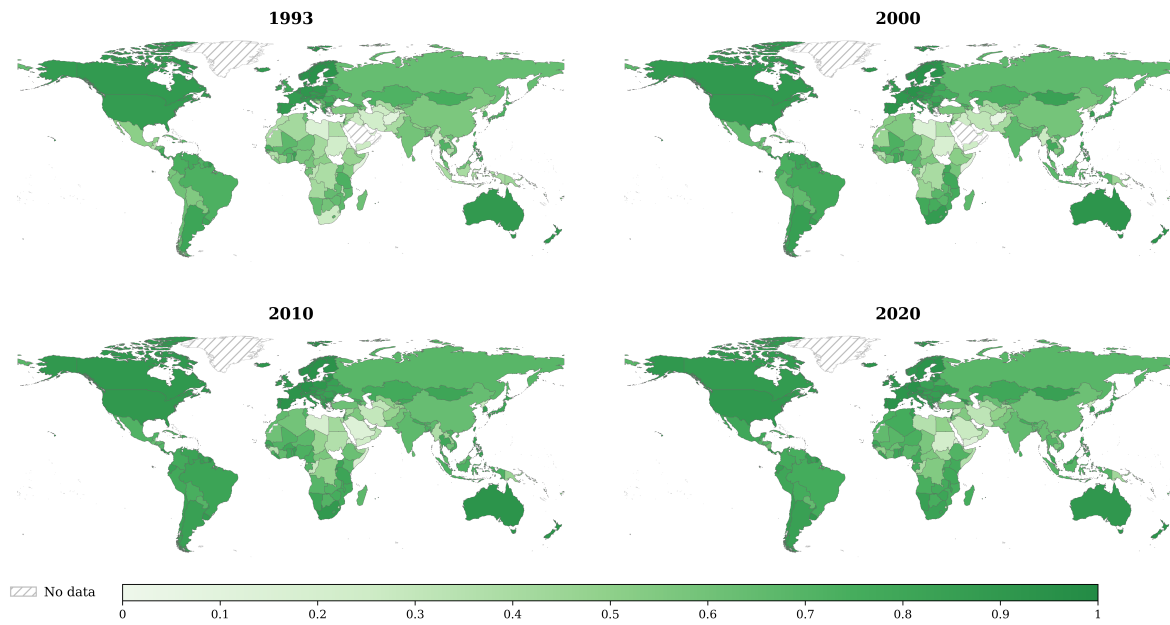


Figure A1: Variation in the Women's Political Empowerment Index across countries, by year.

Table A1: Ordered Logit Estimates- Individual Attitude Variables

Outcome	Without Controls	With controls
Future Environment vs Current Jobs	0.0301 (0.2337)	0.3925*** (0.1029)
Modern Life Harms Environment	0.0369 (0.1069)	0.3332*** (0.1000)
Progress Harms Environment	0.0117 (0.2090)	0.2213** (0.0878)
Personal Environmental Agency	0.0226 (0.1213)	0.1735*** (0.0664)
Willingness to Pay Higher Prices	-0.3050* (0.1706)	-0.1165*** (0.0192)
Willingness to Pay Higher Taxes	-0.2598 (0.2001)	-0.0713*** (0.0186)
Willingness to Accept Lower Living Standards	-0.0967 (0.2000)	-0.0366*** (0.0191)
Costly Environmental Action	-0.0283 (0.1860)	-0.3522*** (0.0812)
Risk of Car Pollution	0.1631 (0.2039)	0.2536*** (0.0886)
Risk of Industrial Air Pollution	0.4427* (0.2468)	0.2062*** (0.0364)
Risk of Agricultural Chemicals	0.3928** (0.1643)	0.1383*** (0.0360)
Risk of Water Pollution	0.4481** (0.2011)	0.1980** (0.0890)
Risk of Global Temperature Rise	0.2492 (0.1823)	0.2563*** (0.0338)
N	138,610	138,610
Pseudo R^2	0.034	0.373

Notes: Column (1) lists the ordinal outcome variables. Column (2) presents estimates without controls; Column (3) includes the full set of controls. Each cell reports the coefficient on the standardised Women's Political Empowerment Index from a separate regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Cluster robust standard errors at the country level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: First Stage Estimates- Gender Quota as Instrument (Summary Indices)

Outcome	Index: Anderson (2008)	Index: PCA	Index: Simple Average
Environmental concern	0.3537*** (0.0567) KPF-38.88	0.3555*** (0.0596) KPF- 35.63	0.3537*** (0.0567) KPF- 38.88
Willingness to pay	0.3567*** (0.0582) KPF- 37.53	0.3578*** (0.0606) KPF- 34.91	0.3567*** (0.0582) KPF- 37.53
Environmental risk perception	0.3540*** (0.0565) KPF- 39.19	0.3493*** (0.0588) KPF- 35.31	0.3540*** (0.0565) KPF- 39.19

Notes: Column (1) lists the outcome domain. Columns (2)–(4) present the first-stage estimates and Kleibergen-Paap F-statistics for each aggregation method. Each row presents estimates for a different outcome domain, using three alternative aggregation methods — Anderson (2008), Principal Component Analysis, and simple averaging. The Environmental Concern Index aggregates three variables: Future Environment vs. Current Jobs, Modern Life Harms Environment, and Progress Harms Environment. The Willingness to Pay Index aggregates three variables: Willingness to Pay Higher Prices, Willingness to Pay Higher Taxes, and Willingness to Accept Lower Living Standards. The Environmental Risk Perception Index aggregates five variables: Risk of Car Pollution, Risk of Industrial Air Pollution, Risk of Agricultural Chemicals, Risk of Water Pollution, and Risk of Global Temperature Rise. Each cell reports the coefficient on the binary gender quota instrument from a separate regression. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: OLS Estimates- Heterogeneity Effects by Gender (Individual Attitude Variables)

Outcome	Male	Female
Future Environment vs Current Jobs	-0.0025 (0.0269)	0.0925*** (0.0251)
Modern Life Harms Environment	0.0633** (0.0277)	0.0542** (0.0257)
Progress Harms Environment	0.0448 (0.0408)	0.1347** (0.0569)
Personal Environmental Agency	0.0591** (0.0269)	0.0711*** (0.0213)
Willingness to Pay Higher Prices	-0.1257*** (0.0293)	-0.1056*** (0.0268)
Willingness to Pay Higher Taxes	-0.1311*** (0.0285)	-0.0855*** (0.0269)
Willingness to Accept Lower Living Standards	-0.1634*** (0.0322)	-0.0283** (0.0113)
Costly Environmental Action	-0.2361*** (0.0415)	-0.0727* (0.0394)
Risk of Car Pollution	0.0912*** (0.0309)	0.0525* (0.0307)
Risk of Industrial Air Pollution	0.0828** (0.0396)	0.1269*** (0.0362)
Risk of Agricultural Chemicals	0.2079*** (0.0395)	0.1514*** (0.0371)
Risk of Water Pollution	0.1058*** (0.0383)	0.0647* (0.0376)
Risk of Global Temperature Rise	0.1429** (0.0594)	0.1432** (0.0570)
N	63,496	74,897
Adjusted R^2	0.376	0.334

Notes: Column (1) lists the binary outcome variables. Column (2) restricts the sample to male individuals; Column (3) restricts the sample to female individuals. Each cell reports the coefficient on the standardised Women's Political Empowerment Index from a separate regression. All specifications include individual and country level controls, country and year fixed effects, and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: OLS Estimates - Heterogeneity Effects by Gender Using Anderson (2008) Summary Indices

Outcome	Male	Female
Environmental Concern	0.1580** (0.0663)	0.2003*** (0.0658)
Willingness to Pay	-0.1934*** (0.0620)	-0.0997* (0.0531)
Environmental risk perception	0.1598** (0.0655)	0.2224*** (0.0577)
N	63,496	74,897
Adjusted R^2	0.253	0.381

Notes: Column (1) lists the outcome domain. Column (2) restricts the sample to male individuals; Column (3) restricts the sample to female individuals. Each row presents estimates for a different outcome domain, using the Anderson (2008) summary index. Each cell reports the coefficient on the standardised Women's Political Empowerment Index from a separate regression. All specifications include individual and country level controls, country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Multiple Hypothesis Correction: BKY sharpened q values

Outcome	Unadjusted p values	BKY sharpened q values
Future Environment vs Current Jobs	0.000	0.001
Modern Life Harms Environment	0.007	0.005
Progress Harms Environment	0.000	0.001
Personal Environmental Agency	0.000	0.001
Willingness to Pay Higher Prices	0.022	0.01
Willingness to Pay Higher Taxes	0.003	0.004
Willingness to Accept Lower Living Standards	0.028	0.011
Costly Environmental Action	0.008	0.005
Risk of Car Pollution	0.000	0.001
Risk of Industrial Air Pollution	0.003	0.004
Risk of Agricultural Chemicals	0.000	0.001
Risk of Water Pollution	0.003	0.004
Risk of Global Temperature Rise	0.017	0.008

Notes: BKY sharpened q-values refers to Benjamini, Krieger, and Yekutieli (2006) sharpened q values. Sharpened q values can sometimes fall below the corresponding unadjusted p-values. This happens because, when many hypotheses are actually rejected, the procedure can tolerate a few false rejections as well without violating the overall FDR bound. Sharpened q-values are calculated using the Stata codes provided by Anderson (2008).

Table A6: Lewbel IV-TSLS estimates (Individual Attitude Variables)

Outcome	With Controls
Future Environment vs Current Jobs	0.2470*** (0.0611)
Modern Life Harms Environment	0.3117*** (0.1142)
Progress Harms Environment	0.2330** (0.0930)
Personal Environmental Agency	0.2844** (0.1311)
Willingness to Pay Higher Prices	-0.4428** (0.2044)
Willingness to Pay Higher Taxes	-0.1260*** (0.0458)
Willingness to Accept Lower Living Standards	-0.1461* (0.0801)
Costly Environmental Action	-0.1920* (0.1121)
Risk of Car Pollution	0.1684*** (0.0543)
Risk of Industrial Air Pollution	0.2363*** (0.0788)
Risk of Agricultural Chemicals	0.3910*** (0.1121)
Risk of Water Pollution	0.2500*** (0.0873)
Risk of Global Temperature Rise	0.2813*** (0.0985)
N	138,610
Adjusted R^2	0.4448
First stage coefficient	0.2452*** (0.0467)
Kleibergen-Paap F statistic	27.57

Notes: Column (1) lists the binary outcome variables. Column (2) restricts the sample to male individuals; Column (3) restricts the sample to female individuals. Each cell reports the second-stage coefficient on the instrumented standardised Women's Political Empowerment Index from a separate TSLS regression. All specifications include country and year fixed effects and are weighted using ISSP survey weights. Wild cluster bootstrap standard errors at the country level (999 replications) are in parentheses. First-stage coefficients and F-statistics are also provided.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Kinky Least Squares Estimates (Individual Attitude Variables)

Variable	Coefficient
Future Environment vs Current Jobs	0.2074*** (0.0111)
Modern Life Harms Environment	0.1784*** (0.0108)
Progress Harms Environment	0.1579*** (0.0101)
Personal Environmental Agency	0.1664*** (0.0111)
Willingness to Pay Higher Prices	-0.2001*** (0.0100)
Willingness to Pay Higher Taxes	-0.1732*** (0.0083)
Willingness to Accept Lower Living Standards	-0.1418*** (0.0087)
Costly Environmental Action	-0.1807*** (0.0084)
Risk of Car Pollution	0.1468*** (0.0089)
Risk of Industrial Air Pollution	0.1129*** (0.0076)
Risk of Agricultural Chemicals	0.1962*** (0.0104)
Risk of Water Pollution	0.1721*** (0.0110)
Risk of Global Temperature Rise	0.1770*** (0.0089)
N	138,610

Notes: Column (1) lists the binary outcome variables. Column (2) presents estimates with the full set of controls. Each cell reports the coefficient on the standardised Women's Political Empowerment Index from a separate regression. The estimation results correspond to the endogeneity correlation -0.01, and are robust across other similar endogeneity correlation values up to -0.1. All specifications include country and year fixed effects. Confidence intervals are computed following Kripfganz and Kiviet's (2021) approach, which considers the union of intervals across a grid of endogeneity correlations to produce asymptotically conservative inference.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.